

Firm Scope and Innovation: The Role of Intangibles^{*}

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Abstract

This paper develops a unified theoretical framework to study how the composition of intangible investment and innovation direction jointly shape long-run growth and creative destruction. I build an endogenous growth model in which firms choose between vertical and horizontal innovation and allocate investment across R&D and firm-specific intangibles (brand value and organizational capital). Using firm-level data, I discipline the model and find that productivity, markups, and intangible investment composition vary systematically with firm scope: firms with narrower scope concentrate investment in knowledge capital, whereas firms with broader scope allocate relatively more to brand value and organizational capital. The calibrated model shows that firm scope determines intangible composition, which governs innovation direction, and that growing scope and organizational capital jointly raise entry barriers and suppress creative destruction. Size-based policies misrepresent firms' investment incentives by treating vertical and horizontal expansion symmetrically, whereas scope-dependent policies targeting intangible-driven externalities lead to higher rates of creative destruction and innovation.

Keywords: Schumpeterian growth, step-by-step innovation, intangibles, firm dynamics, span of control.

JEL Classification: E22, O31, O32, O33, O34.

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1. Introduction

Horizontal and vertical innovation are two core strategies in the modern endogenous growth literature for generating long-run growth through creative destruction. Firms can engage in horizontal innovation by expanding their product portfolios (Klette and Kortum, 2004) or pursue vertical innovation by improving the quality of existing products (Aghion and Howitt, 1992; Aghion *et al.*, 2001). At the same time, literature also documents that firms invest in intangibles not only to innovate but also to build intangible assets such as brand value and organizational capital that strengthen market positions rather than drive innovation-led growth (Cavenaile and Roldan-Blanco, 2021; Crouzet and Eberly, 2019).

While the directions of innovation and composition of intangibles are well documented separately, understanding their interaction is crucial for shaping effective policies. To be concrete, Apple is a vertically integrated firm focused on mobile communication technologies and invests heavily in R&D. By contrast, 3M is active in transportation, safety, consumer, and health care industries, and although R&D remains its largest intangible investment, it allocates a substantially larger share toward brand and organizational capital relative to Apple¹. This contrast in firm scope shows weakness of flat and size-based policies. By treating Apple and 3M symmetrically, such policies can either penalize firms that invest heavily in R&D or equally reward those with excessively accumulated firm-specific intangibles, regardless of their aggregate consequences. Yet existing theories provide no framework for distinguishing these two firms' investment incentives or for designing policies that account for this heterogeneity.

What are the aggregate consequences of the interaction between innovation direction and intangible composition for economic growth and creative destruction? To answer this question, I proceed in two steps. First, I develop a unified theoretical framework in which firms engage in both vertical and horizontal innovation and allocate investment across different types of intangibles. Second, I document a new set of empirical regular-

¹The measurement details are given in Section 3.2.

ities showing how firm-level outcomes and investment choices vary systematically with firm scope, which I use to discipline the model. The calibrated model shows that firms operating with a narrower scope concentrate investment in knowledge capital, whereas firms with a broader scope allocate relatively more to brand value and organizational capital, and the composition of intangible types in turn determines firms' innovation direction. This joint determination of innovation direction and intangible composition implies that policies conditioned solely on firm size are insufficient, as firm size contains vertical and horizontal dimensions and therefore misrepresents firms' intangible investment incentives. By contrast, scope-dependent policies that target the externalities associated with intangible heterogeneity lead to higher rates of creative destruction and innovation.

To formalize intangible heterogeneity, I classify intangible assets according to their spillover effects across firms. In this paper, transferable intangibles consist of knowledge capital² with the associated R&D investments. These assets can be transferred between firms and generate spillover effects due to their non-rivalrous and partially excludable nature (Romer, 1990). In contrast, embedded intangibles comprise brand value³ and organizational capital⁴ are inherently firm-specific and inseparable from the firm that created them and do not generate spillovers. Consequently, when a firm exits the market, the economic value of embedded intangibles becomes a sunk cost.

In Section 2, I develop an endogenous growth model that the economy consists of a single final-good sector and a continuum of intermediate-good sectors. In each intermediate goods sector, a single superstar firm and a continuum of fringe firms produce intermediate goods for the final goods sector with a la Bertrand competition. Superstar

²Following Griliches (1979), R&D investment generates a stock of knowledge capital that accumulates over time. Examples of such knowledge include patents and innovation-related software, which illustrate the types of assets that represent technological know-how.

³Brand value is a demand shifter, positively influencing the perceived quality of a firm's output (Cavenaile and Roldan-Blanco, 2021; Cavenaile *et al.*, 2025a). Evidence also suggests that brand value increases consumer awareness through targeted marketing (Cavenaile *et al.*, 2025b; Baslandze *et al.*, 2023)

⁴Organizational capital conceptualized as managerial productivity, including the firm's embodied managerial talent and its contribution to future production profitability (Carlin *et al.*, 2012; Eisefeldt and Panikolaou, 2013; Prescott and Visscher, 1980)

firms can invest in embedded intangibles, which improve managerial productivity and the perceived quality of their products through brand value and organizational capital. Alternatively, they can invest in transferable intangibles, which enhance product quality in two ways: (i) by vertically improving existing products and (ii) by expanding their scope through entry into new sectors. Fringe firms produce homogeneous products and cannot accumulate brand value and organizational capital. When a superstar firm exits a sector, its transferable intangibles—the established product quality in that sector—become freely available to fringe firms. For fringe firms, the only path to becoming a superstar is through radical innovation.

The model incorporates two key frictions. First, two types of intangibles differ in their scalability. Embedded intangibles generate economies of scope, as their benefits increase when a firm is active in more sectors because they are fully mobile across sectors, whereas transferable intangibles require separate investment for each sector. Second, while vertical innovation imposes no span-of-control constraints, expanding scope divides managerial attention across sectors, therefore reducing managerial quality per sector (Lucas, 1978)⁵. However, when a superstar firm expands into more sectors, its total managerial quality increases, so that while managerial quality per sector declines with span-of-control constraint, the superstar’s total managerial capacity increases with scope. Moreover, superstar firm could also strengthen their managerial quality through the accumulation of organizational capital. Therefore, when facing a competitive threat, the incumbent superstar leverages both its embedded intangibles and scope, creating entry barriers for challengers by deploying its total managerial capacity.

In Section 3, I combine firm-level data from Compustat with the product-market fluidity dataset of Hoberg *et al.* (2014) to document how markups, productivity, intangible investment, and competitive pressure vary systematically with firm scope. To isolate causal direction, I employ the patent-value dataset from (Kogan *et al.*, 2017) as an innovation shock to compare the responses of single- and multi-sector firms using local

⁵See also Jovanovic (2025). Alternatively, Acemoglu *et al.* (2018) propose that when a firm expands its scope, more skilled labor needs to be allocated to operational activities.

projections. Single-sector firms show stronger reallocation toward transferable intangibles, larger productivity gains, higher markup responses, and greater competitive pressure from rivals compared to multi-sector firms facing an equivalent shock.

In Section 4, I calibrate the model to firm-level cross-sectional moments and derive three quantitative predictions. First, scope exerts a first-order effect on innovation direction: as firms expand into additional sectors, the span-of-control constraint erodes market share in each individual sector, shifting incentives away from transferable and toward embedded intangible investment. Second, even firms of similar scope pursue sharply different innovation strategies depending on the composition of their intangible capital, generating rich nonlinear heterogeneity that standard models neglect. Third, as scope and embedded intangibles grow jointly, the incumbent’s managerial capacity rises, raising the displacement threshold for entrants and suppressing creative destruction. These three forces together drive a wedge between private and social returns to innovation: broader-scope firms investing largely in embedded intangibles erect larger entry barriers and generate smaller knowledge spillovers, while focused firms concentrating in transferable capital contribute more to creative destruction and aggregate productivity growth.

In Section 5, I use the calibrated model to quantify the aggregate consequences of embedded intangible accumulation and firm scope. The embedded intangible channel emerges as the dominant source of misallocation, as broad-scope firms erect entry barriers that suppress creative destruction, yet their removal erodes the innovation rents sustaining long-run productivity growth. These findings suggest that the social cost of broad-scope superstar firms operates primarily through the embedded intangible channel rather than scope expansion per se, pointing to targeted policy interventions rather than blanket restrictions on firm size

In Section 6, I evaluate three policy instruments targeting the misallocation identified in the counterfactual analysis. A welfare-maximizing design that subsidizes single-sector superstar firms, taxes broad-scope incumbents, and complements both with a fringe entrant subsidy achieves simultaneous consumption and growth gains, whereas instruments

targeting long-run growth alone impose large upfront consumption costs. The fringe subsidy proves indispensable, as restoring competitive pressure at the entry margin generates disproportionate aggregate gains that incumbent-facing tax instruments alone cannot replicate.

Related Literature. A growing literature documents that intangible investment increases market concentration and markups ([Chiavari and Goraya, 2025](#); [Crouzet and Eberly, 2019](#); [Weiss, 2020](#)). [De Ridder \(2024\)](#) treats software as a firm-specific fixed cost and shows that firms with higher software capability strategically increase their fixed-cost investment to lower their marginal production costs, which creates entry barriers. [Aghion et al. \(2023\)](#) distinguish between product and process innovation and they document that declining overhead costs or rising productivity efficiency are the primary forces behind increasing market concentration. In both frameworks, intangible capabilities are treated as a firm's inborn characteristics rather than endogenous investment choices. [Cavenaile and Roldan-Blanco \(2021\)](#) and [Cavenaile et al. \(2025a\)](#) show that advertising can substitute for R&D and dampen innovation intensity, a finding further extended by [Pearce and Wu \(2024\)](#), who show that brand value can be transferred in the secondary market rather than only accumulated within the firm. This paper contributes to the literature by establishing that intangible composition is an endogenous determinant of innovation direction and evolves endogenously with firm scope.

The literature has predominantly explained firms' choice of innovation direction through differences in firm size. [Akcigit and Kerr \(2018\)](#) show that smaller firms are more likely to engage in external innovation while larger firms tend toward quality-improving vertical innovation. [Berlingieri et al. \(2025\)](#) documents that firms expand their size by adding a series of products simultaneously in the early period, while [Garcia-Macia et al. \(2019\)](#) finds that most within-firm innovation arises from incumbents improving existing products rather than entering new lines. I show that intangible composition is an important channel to innovation direction that operates independently of firm size. Firms can choose different innovation paths depending on their allocation of investment across transferable and embedded intangible assets, which firm size alone cannot account for the observed

heterogeneity in innovation strategies.

This paper provides a complementary explanation for documented trends, including declining knowledge spillovers ([Akcigit and Ates, 2021](#); [Akcigit and Ates, 2023](#)), lower patent quality ([Olmstead-Rumsey, 2019](#)), production lock-in ([Casal, 2024](#)), and rising entry barriers ([Gutiérrez and Philippon, 2019](#)). As the firm's scope expands, investment shifts from transferable toward embedded intangibles, depressing knowledge spillovers and raising entry barriers.

This paper also contributes to the misallocation literature ([Hsieh and Klenow, 2009](#); [Restuccia and Rogerson, 2008](#)) by identifying two distinct channels through which intangible heterogeneity generates allocative distortions. The first operates through markup dispersion ([Peters, 2020](#); [Edmond *et al.*, 2023](#)), driven by the accumulation of both transferable and embedded intangibles. The second operates through entry barriers: the joint expansion of embedded intangibles and firm scope makes creative destruction asymmetric, generating persistent misallocation that markup-based measures substantially understate.

Finally, the literature on span-of-control has focused primarily on hierarchical organization, knowledge-flow frictions, and managerial ability ([Smeets *et al.*, 2019](#); [Bandiera *et al.*, 2014](#); [Garicano, 2000](#); [Bloom and Van Reenen, 2007](#)) and with ([Jovanovic, 2025](#)) linking span-of-control constraints to the firm-size distribution at the macro level. I contribute to this literature by disaggregating firm size into vertical and horizontal dimensions of innovation and showing that span-of-control frictions arise only from horizontal expansion. This distinction isolates how scope-driven managerial constraints shape intangible investment, innovation direction, and aggregate growth.

2. Theoretical Model

The theoretical framework builds on the step-by-step vertical innovation framework from [Akcigit and Ates \(2023\)](#) and on the vertical and horizontal innovation frameworks from [Peters \(2020\)](#). I combine both frameworks by introducing two distinct types of intangible

investment and span-of-control frictions that arise exclusively from expanding scope.

2.1. Economic Environment

Time is continuous and indexed by t . Household preferences are described by a logarithmic utility function

$$\int_0^\infty e^{-\rho t} \ln(C_t) dt, \quad (1)$$

where C_t denotes household consumption and $\rho > 0$ is the time discount rate. The household budget constraint is

$$\dot{A}_t = r_t A_t + w_t - C_t, \quad (2)$$

with A_t denotes total household assets, w_t is the wage rate, and r_t is the interest rate. Labor is supplied inelastically, and the price of the consumption good is normalized to 1, where w_t and r_t are expressed in units of the consumption good. Since households own all firms in the economy, total assets equal the aggregate value of superstar and fringe firms across all intermediate-good sectors

$$A_t = \int_0^1 (V_{sjt} + V_{fjt}) dj,$$

with V_{sjt} and V_{fjt} denoting the values of the superstar and fringe firms.

The final good is produced by aggregating a continuum of intermediate varieties $j \in [0, 1]$ according to

$$\ln(Y_t) = \int_0^1 \ln(y_{jt}) dj, \quad (3)$$

in a perfectly competitive market. In each intermediate-good sector, a single superstar firm and a continuum of fringe firms compete à la Bertrand to supply the final good producer. A superstar firm s operates across a set of sectors $J_s \subseteq [0, 1]$, with $n_s = |J_s| \in \mathbb{Z}_+$ denoting the number of sectors in which it holds the leading technology. Output in each sector is aggregated with a constant elasticity of substitution technology

$$y_{jt} = (\Xi(e_{st}) y_{sjt}^\varepsilon + (1 - \Xi(e_{st})) y_{fjt}^\varepsilon)^{\frac{1}{\varepsilon}} \quad (4)$$

where $\Xi(e_{st}) \equiv \frac{\chi(e_{st})}{1 + \chi(e_{st})}$ denotes perceived quality of superstar firms, and $\varepsilon \in (0, 1)$. In each sector, the superstar firm produces a differentiated good whose perceived quality is increasing in its brand value. Specifically, $\chi(e_{st}) = (\xi e_{st})^\beta$ is an endogenous and concave demand shifter, where e_{st} denotes the embedded intangibles of superstar firm s , $\xi \in (0, 1)$ is the share of embedded intangibles associated with brand value, and $\beta \in (0, 1)$ governs the curvature of the demand shifter. The term $\Xi(e_{st})$ captures the notion that a higher stock of brand value raises the perceived quality of the superstar's product relative to the fringe. For simplicity, the brand value of fringe firms is normalized to one, and they produce a homogeneous good aggregated as

$$y_{fjt} = \int_0^1 y_{ijt} di, \quad (5)$$

where each fringe firm $i \in (0, 1)$ takes the market price as given.

The production function of superstar firm s in sector j at time t is

$$y_{sjt} = \underbrace{q_{sjt}}_{\text{Product Quality}} \underbrace{\psi(e_{st}, n_{st})}_{\text{Managerial Productivity}} \underbrace{l_{sjt}}_{\text{Labor Input}}, \quad (6)$$

where q_{sjt} denotes product quality, l_{sjt} is labor input, and $\psi(e_{st}, n_{st})$ captures managerial productivity, defined as

$$\psi(e_{st}, n_{st}) = \frac{((1 - \xi) e_{st})^\alpha}{\gamma n_{st}^\alpha}.$$

The component $(1 - \xi)e_{st}$ represents the share of embedded intangibles allocated to organizational capital, which governs managerial efficiency. The variable n_{st} denotes the number of sectors operated by firm s at time t . As firm scope n_{st} increases, managerial productivity per sector declines due to the span-of-control friction: the firm's managerial attention is distributed across a larger number of sectors, reducing the managerial capacity allocated to each individual sector. The parameter α controls the curvature of both organizational capital and the span-of-control constraint, while γ is a scale parameter. Aggregating the managerial productivity of superstar firm s across all active sectors

generates total managerial capacity⁶

$$\Lambda \cdot e_{st}^\alpha \cdot n_{st}^{1-\alpha}, \quad \text{with} \quad \Lambda \equiv \frac{(1 - \xi)^\alpha}{\gamma}.$$

When $0 < \alpha < 1$, total managerial capacity increases with both organizational capital and firm scope, yet each exhibits diminishing marginal returns.⁷ Total managerial capacity determines the incumbent superstar's ability to defend its market position against challenger entry, a mechanism formalized in the creative destruction section.

Fringe firms produce output using a linear technology

$$y_{fjt} = q_{fjt} l_{fjt}, \tag{7}$$

where q_{fjt} denotes product quality and l_{fjt} is labor input. They operate in a single sector, cannot accumulate embedded intangibles, and have managerial quality normalized to one. Their product quality is inherited: when a superstar firm exits a sector, its transferable intangibles become publicly available and fringe firms adopt the previous superstar's quality level.

Superstar firms pursue three distinct innovation strategies. They can invest in transferable intangibles to either improve the quality of an existing product through vertical innovation or expand into a new sector through horizontal innovation. Additionally, they can invest in embedded intangibles to raise their brand value and organizational capital across all sectors in which they operate (see [Figure 1](#)). The first two strategies require sector-specific investment because product quality improvements embody knowledge specific to each sector's technology. By contrast, embedded intangible investment is firm-specific: an improvement in brand value or organizational capital simultaneously raises the embedded intangible stock of all sectors within the firm's portfolio, reflecting

⁶Managerial productivity is symmetric across sectors, so total managerial capacity equals per-sector managerial productivity $\psi(e_{st}, n_{st})$ multiplied by the number of active sectors n_{st} .

⁷When $\alpha > 1$, diseconomies of scope emerge: coordination costs outweigh the benefits of expansion, and total managerial capacity decreases with firm scope.

the economies of scope that embedded intangibles generate.

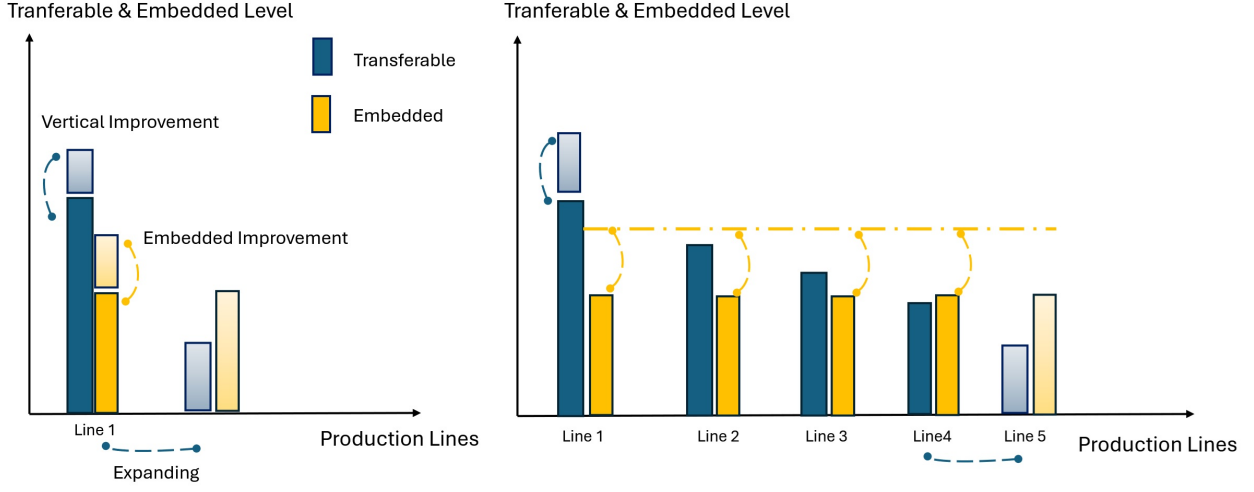


Figure 1. Firm Investment and Innovation Types

The variables $I_{s jt}^{\text{Emb}}$, $I_{s jt}^{\text{Ver}}$, and $I_{s jt}^{\text{Hor}}$ denote the investment of superstar firm s in embedded intangibles, vertical innovation in an existing sector, and horizontal innovation into a new sector, respectively. Each unit of investment generates a successful flow rate of innovation: z_{jt}^{Emb} for embedded intangibles, z_{jt}^{Ver} for vertical, and z_{jt}^{Hor} for horizontal innovation investment. Investments are subject to convex costs:

$$I_{s jt}^{\text{Ver}} = \gamma^{\text{Ver}} (z_{s jt}^{\text{Ver}})^{\vartheta^{\text{Ver}}} Y_t, \quad I_{s jt}^{\text{Hor}} = \gamma^{\text{Hor}} (z_{s jt}^{\text{Hor}})^{\vartheta^{\text{Hor}}} Y_t, \quad (8)$$

$$\text{and} \quad I_{s jt}^{\text{Emb}} = \gamma^{\text{Emb}} (z_{s jt}^{\text{Emb}})^{\vartheta^{\text{Emb}}} Y_t,$$

where the cost parameters γ^{Ver} , γ^{Hor} , and γ^{Emb} govern the scale of each investment cost function, ϑ^{Ver} , ϑ^{Hor} , and ϑ^{Emb} determine their curvature, and all costs scale with aggregate output Y_t .

Fringe firms invest only in transferable intangibles within their sector, targeting radical innovations that would allow them to displace the incumbent superstar. Their investment cost function takes the form

$$I_{f jt} = \gamma^f (z_{f jt})^{\vartheta^f} Y_t, \quad (9)$$

where γ^f is the cost scale parameter and ϑ^f determines the curvature of investment.

Taken together, the total investment in transferable intangibles in sector j at time t combines vertical and horizontal investments by the incumbent superstar and investment by fringe firms:

$$I_{jt}^T = I_{sjt}^{\text{Ver}} + I_{sjt}^{\text{Hor}} + I_{fjt}. \quad (10)$$

A successful quality improvement — whether vertical within an existing sector or horizontal into a new sector — improves product quality by a factor of $\lambda > 1$, while a successful embedded intangible investment raises brand value and organizational capital by a factor of $\theta > 1$ across all active sectors simultaneously. Product quality and embedded intangibles therefore evolve as

$$q_{sjt} = \lambda^{m_{sjt}} q_{sj0}, \quad \text{and} \quad e_{st} = \theta^{k_{st}} e_{s0}, \quad (11)$$

where m_{sjt} denotes the cumulative number of product-quality innovations by firm s in sector j up to time t , and k_{st} denotes the cumulative number of embedded intangible innovations by firm s up to time t , with initial levels normalized to $q_{sj0} = 1$ and $e_{s0} = 1$.

The quality gap between the superstar and the fringe in sector j at time t is therefore

$$\frac{q_{sjt}}{q_{fjt}} = \lambda^{m_{sjt} - m_{fjt}} = \lambda^{m_{jt}}, \quad (12)$$

where $m_{jt} \equiv m_{sjt} - m_{fjt}$ denotes the transferable intangible gap. Since the embedded intangible stock of fringe firms is normalized to one, the embedded intangible gap simplifies to the superstar's own stock:

$$\frac{e_{st}}{e_{ft}} = \frac{\theta^{k_{st}}}{1} = \theta^{k_{st}}. \quad (13)$$

Assumption (Managerial Reallocation Capability). *When a superstar firm — whether incumbent or challenger — faces competition, it can temporarily concentrate its entire managerial capacity on the contested sector. This reallocation is costless in the short run and determines the outcome of the price competition stage.*

Creative destruction arises either from a challenger superstar firm expanding its scope or from a fringe firm pursuing radical innovation. When a superstar firm successfully pursues horizontal innovation, it enters a randomly selected sector j' , improving product quality by a factor of λ such that $q_{sjt} = \lambda q_{s'jt}$, and displaces the incumbent if its effective unit cost — defined as the wage relative to total managerial capacity — falls below that of the incumbent:

$$\frac{w_t}{\Lambda q_{sjt} e_{st}^\alpha n_{st}^{1-\alpha}} < \frac{w_t}{\Lambda q_{s'jt} e_{s't}^\alpha n_{s't}^{1-\alpha}}.$$

The probability that the entrant successfully displaces the incumbent is

$$\mathbb{P}_{s>s'} \equiv \sum_{m_{s'}=1}^{\bar{m}} \sum_{k_{s'}=1}^{\bar{k}} \sum_{n_{s'}=1}^{\bar{n}} \mathbb{I}\{\lambda (\theta^{k_s})^\alpha n_{st}^{1-\alpha} > (\theta^{k_{s'}})^\alpha n_{s't}^{1-\alpha}\} \mu_t(m_{s'}, k_{s'}, n_{s'}),$$

where s denotes the entrant and s' the incumbent, with the left-hand side reflecting the entrant's total managerial capacity after the quality improvement and the right-hand side the incumbent's. Here $\mu_t(m_{s'}, k_{s'}, n_{s'}) \in [0, 1]$ denotes the mass of superstar firms in state $(m_{s'}, k_{s'}, n_{s'})$ at time t . The state space is finite and discrete, given by $\mathcal{M} \times \mathcal{K} \times \mathcal{N}$, with $\mathcal{M} = \{1, \dots, \bar{m}\}$, $\mathcal{K} = \{1, \dots, \bar{k}\}$, and $\mathcal{N} = \{1, \dots, \bar{n}\}$ with satisfies

$$\sum_{m_{s'}, k_{s'}, n_{s'}} \mu_t(m_{s'}, k_{s'}, n_{s'}) = 1. \quad (14)$$

The probability of losing the specific sector j' is given by

$$\mathbb{P}_{c>s'} = \sum_{m_c, k_c, n_c} \mathbb{I}\{\lambda (\theta^{k_c})^\alpha n_{ct}^{1-\alpha} > (\theta^{k_s})^\alpha n_{st}^{1-\alpha}\} \mu_t(m_c, k_c, n_c).$$

Fringe firms cannot accumulate embedded intangibles and therefore cannot compete with the incumbent on managerial capacity. Their only way to be a superstar is through radical innovation, which requires jumping multiple steps.⁸ on the quality ladder by a

⁸Fringe firms in the model correspond to small firms in the data. A vast literature documents that small and young firms grow faster than their larger counterparts (see [Evans, 1987](#), [Haltiwanger et al., 2013](#)).

factor $\lambda^{\bar{m}} > \lambda$, such that $q_{s jt} = \lambda^{\bar{m}} q_{s' jt}$ ⁹ Since fringe firms are symmetric and invest identically within each sector, a fringe firm successfully displaces the incumbent in sector j if and only if

$$\mathbb{I}_{f>s'} \equiv \mathbb{I}\{\lambda^{\bar{m}} > (\theta^{k_{s'}})^{\alpha} n_{s'}^{1-\alpha}\},$$

where the right-hand side depends solely on the incumbent's state $(k_{s'}, n_{s'})$. Entry conditions through creative destruction show that the barrier is increasing in both the incumbent's embedded intangible stock $\theta^{k_{s'}}$ and its scope $n_{s'}$, regardless of whether the challenger is a fringe firm or superstar. Moreover, two embedded intangibles play asymmetric roles in the entry condition. Since only organizational capital enters total managerial capacity, eliminating it reduces entry barriers. By contrast, eliminating brand value raises entry barriers, as any increase in the embedded intangible stock then translates one-to-one into a stronger competitive advantage against challengers.

2.2. Equilibrium

The equilibrium has a static and a dynamic component. The analysis begins with the static equilibrium, determining prices and allocations for a given set of states. I then define the Markov Perfect Equilibrium for the dynamic game, outlining the value functions, optimal policy functions, and the evolution of the aggregate state distribution.

A household maximizes utility subject to the budget constraint, yielding the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho. \quad (15)$$

Along the balanced growth path, consumption and output grow at the same rate $g = r - \rho$, and the transversality condition holds. The final-good producer's demand for intermediate goods in sector j satisfies

$$p_{jt} = \frac{Y_t}{y_{jt}}. \quad (16)$$

⁹Following [Akcigit et al. \(2026\)](#), drastic innovation by fringe firms can alternatively be modeled using a probability distribution $\mathbb{F}(\cdot)$ over the innovation jumps, rather than a fixed step of size $\bar{m}$.

This implies the following demand functions for the superstar and fringe firms:

$$y_{sjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t (\Xi(e_s))^{\frac{1}{1-\varepsilon}} \quad \text{and} \quad y_{fjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{fjt}^{\frac{1}{\varepsilon-1}} Y_t (1 - \Xi(e_s))^{\frac{1}{1-\varepsilon}}, \quad (17)$$

where p_{sjt} and p_{fjt} are the prices charged by the superstar and fringe firms, respectively. The corresponding ideal price index for sector j is given by

$$p_{jt} = \left((\Xi(e_s))^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s))^{\frac{-1}{\varepsilon-1}} p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{\frac{\varepsilon-1}{\varepsilon}}. \quad (18)$$

The Cobb-Douglas aggregator implies equal expenditure shares across all sectors. The market share of superstar firm s in sector j at time t is

$$\frac{p_{sjt} y_{sjt}}{p_{jt} y_{jt}} = \frac{p_{sjt} y_{sjt}}{Y_t} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \Xi(e_s)^{\frac{1}{1-\varepsilon}} \equiv \phi_{sjt}, \quad (19)$$

and since market shares sum to one, the fringe firms' share is $1 - \phi_{sjt}$. The equilibrium price of the superstar firm under Bertrand competition is¹⁰

$$p_{sjt} = \frac{1 - \varepsilon \phi_{sjt}}{(1 - \phi_{sjt}) \varepsilon} \underbrace{\frac{w_t n_{st}^\alpha}{\Lambda q_{sjt} e_{st}^\alpha}}_{MC_{sjt}}, \quad (20)$$

where MC_{sjt} denotes the marginal cost of the superstar firm.¹¹ The equilibrium price increases with product quality and the embedded intangible stock, and decreases with the number of sectors operated. The price ratio of fringe to superstar firms is

$$\frac{p_{fjt}}{p_{sjt}} = \frac{(1 - \phi_{sjt}) \varepsilon}{1 - \varepsilon \phi_{sjt}} \cdot \lambda^{m_j} \left(\frac{(1 - \xi) e_{st}}{n_{st}} \right)^\alpha. \quad (21)$$

Substituting the ideal price index from equation (18) into the market share definition in

¹⁰See Appendix B.3 for the full derivations. For the à la Cournot competition, see Appendix B.4.

¹¹Although fringe firms lack independent pricing power, their presence creates a competitive constraint that limits the superstar firm's pricing power and prevents it from extracting monopolistic rents.

equation (19), the superstar's market share can be expressed in terms of relative prices as

$$\phi_{s jt} = \frac{1}{1 + \left(\frac{1}{(\chi(e_{st}))^{\frac{1}{1-\varepsilon}}} \left(\frac{p_{f jt}}{p_{s jt}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \right)}, \quad (22)$$

which depends on the quality gap m_j , the embedded intangible stock e_{st} , and the number of sectors operated n_{st} .

The operational profit and markup of the superstar firm are proportional to its market share:

$$\pi_{s jt} = \frac{(1-\varepsilon)\phi_{s jt}}{1-\varepsilon\phi_{s jt}} Y_t \quad \text{and} \quad \sigma_{s jt} = \frac{1-\varepsilon\phi_{s jt}}{(1-\phi_{s jt})\varepsilon}. \quad (23)$$

Both profit and markup increase with market share and therefore decrease with firm scope, reflecting the diminishing managerial productivity that arises from the span-of-control friction. The optimal labor inputs for superstar and fringe firms are

$$l_{s jt} = \frac{\phi_{s jt}}{\sigma_{s jt}} \omega_t^{-1} \quad \text{and} \quad l_{f jt} = (1-\phi_{s jt}) \omega_t^{-1}, \quad (24)$$

where $\omega_t = w_t/Y_t$ denotes the wage share of the economy. The static equilibrium provides only an implicit solution. Nonetheless, the model yields tractable dynamics because the equilibrium outcomes for both superstar and fringe firms depend solely on their market share, which is determined by the quality gap, the embedded intangible stock, and the number of sectors in which it operates.

The value function of superstar firm s , which depends on the quality gap vector $\mathbf{m} = \{m_j\}_{j=1}^n$, the embedded intangible level k , and the number of sectors operated n .¹² The

¹²To simplify notation, the subscript s is omitted whenever it is clear from context.

firm chooses innovation flow rates z_j^{Ver} , z_j^{Hor} , and z_j^{Emb} to maximize

$$\begin{aligned}
r_t V_t(\mathbf{m}, k, n) - \dot{V}_t(\mathbf{m}, k, n) = & \max_{z_{jt}^{\text{Ver}}, z_{jt}^{\text{Hor}}, z_{jt}^{\text{Emb}}} \sum_{j=1}^n \left(\frac{(1-\varepsilon) \phi_t(m_j, k, n)}{1-\varepsilon \phi_t(m_j, k, n)} Y_t \right. \\
& + \underbrace{\mathbb{P}_{s \geq s'} z_{jt}^{\text{Hor}} [V_{jt}((m_j, 1), k, n+1) - V_{jt}(m_j, k, n)]}_{\text{Horizontal Innovation}} + \underbrace{z_{jt}^{\text{Ver}} [V_{jt}(m_j+1, k, n) - V_{jt}(m_j, k, n)]}_{\text{Vertical Innovation}} \\
& + \underbrace{z_{jt}^{\text{Emb}} [V_{jt}(\mathbf{m}, k+1, n) - V_{jt}(\mathbf{m}, k, n)]}_{\text{Embedded Innovation}} + \underbrace{Z_{c>st}^{\text{Hor}} [-V_{jt}(m_j, k, n)]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s} Z_{jt}^f [-V_{jt}(m_j, k, n)]}_{\text{Fringe Entry}} \\
& \left. - \gamma^{\text{Ver}} (z_{jt}^{\text{Ver}})^{\vartheta^{\text{Ver}}} Y_t - \gamma^{\text{Emb}} (z_{jt}^{\text{Emb}})^{\vartheta^{\text{Emb}}} Y_t - \gamma^{\text{Hor}} (z_{jt}^{\text{Hor}})^{\vartheta^{\text{Hor}}} Y_t \right). \tag{25}
\end{aligned}$$

The left-hand side represents the flow return on the firm's value. The first term on the right-hand side is the operational profit of the superstar firm in sector j . The second and third terms capture the value gains from horizontal and vertical innovation, respectively: a successful horizontal innovation expands the firm's scope from n to $n+1$, while a successful vertical innovation advances the quality gap in sector j by one rung. The fourth term captures the superstar gain from a successful embedded intangible investment, which raises the embedded gap k by one rung across all active sectors simultaneously. The fifth and sixth terms capture the value loss upon displacement by a challenger superstar or a fringe firm. The final three terms are the convex investment costs defined in equation (8). The terms $Z_{c>st}^{\text{Hor}}$ and Z_{jt}^f denote the aggregate innovation rates of challenger superstar firms and fringe firms, respectively. The aggregate horizontal innovation rate of challenger superstars that successfully displace the incumbent is

$$Z_{c>st}^{\text{Hor}} = \sum_{m_c, k_c, n_c} \mathbb{I}\{\lambda (\theta^{k_c})^\alpha n_{ct}^{1-\alpha} > (\theta^{k_s})^\alpha n_{st}^{1-\alpha}\} z_t^{\text{Hor}}(m_c, k_c, n_c) \mu_t(m_c, k_c, n_c),$$

and the aggregate fringe firm innovation rate is $Z_{jt}^f = \int_0^1 z_{ijt} di$.

Along the balanced growth path, aggregate output Y , consumption C , and the value function $V(\mathbf{m}, k, n)$ all grow at the constant rate g . Dividing the value function by aggregate output and defining $v(\mathbf{m}, k, n) \equiv V(\mathbf{m}, k, n)/Y$ yields the normalized superstar value

function, which in the stationary equilibrium satisfies

$$\begin{aligned}
\rho v(\mathbf{m}, k, n) = & \max_{z_j^{\text{Ver}}, z_j^{\text{Hor}}, z_j^{\text{Emb}}} \sum_{j=1}^n \left(\frac{(1-\varepsilon) \phi(m_j, k, n)}{1-\varepsilon \phi(m_j, k, n)} \right. \\
& + \underbrace{\mathbb{P}_{s \geq s'} z_j^{\text{Hor}} \left[v((m_j, 1), k, n+1) - v(m_j, k, n) \right]}_{\text{Horizontal Innovation}} + \underbrace{z_j^{\text{Ver}} \left[v(m_j+1, k, n) - v(m_j, k, n) \right]}_{\text{Vertical Innovation}} \\
& + \underbrace{z_j^{\text{Emb}} \left[v(\mathbf{m}, k+1, n) - v(\mathbf{m}, k, n) \right]}_{\text{Embedded Innovation}} + \underbrace{Z_{c>s}^{\text{Hor}} \left[-v(m_j, k, n) \right]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s} Z_j^f \left[-v(m_j, k, n) \right]}_{\text{Fringe Entry}} \\
& \left. - \gamma^{\text{Ver}} (z_j^{\text{Ver}})^{\vartheta^{\text{Ver}}} - \gamma^{\text{Emb}} (z_j^{\text{Emb}})^{\vartheta^{\text{Emb}}} - \gamma^{\text{Hor}} (z_j^{\text{Hor}})^{\vartheta^{\text{Hor}}} \right). \tag{26}
\end{aligned}$$

Since fringe firms produce a homogeneous good, they earn zero profit in equilibrium and derive value solely from the possibility of displacing the incumbent through radical innovation. The fringe firm chooses innovation intensity z_f to maximize this value, trading off investment costs against the gain from becoming the new superstar. The normalized fringe firm value function $v_f(m, k, n) \equiv V_{ft}(m, k, n)/Y$ in the stationary equilibrium satisfies

$$\begin{aligned}
\rho v_f(m, k, n) = & \max_{z_f} \left\{ \underbrace{\mathbb{I}_{f>s'} z_f \left[v(\bar{m}, 1, 1) - v_f(m, k, n) \right]}_{\text{Radical Innovation}} - \gamma^f (z_f)^{\vartheta^f} \right. \\
& + \underbrace{\mathbb{P}_{s \geq s'} z_j^{\text{Hor}} \left[v_f((m, 1), k, n+1) - v_f(m, k, n) \right]}_{\text{Incumbent Horizontal Innovation}} + \underbrace{z_j^{\text{Ver}} \left[v_f(m+1, k, n) - v_f(m, k, n) \right]}_{\text{Incumbent Vertical Innovation}} \\
& + \underbrace{z_j^{\text{Emb}} \left[v_f(m, k+1, n) - v_f(m, k, n) \right]}_{\text{Incumbent Embedded Innovation}} + \underbrace{\mathbb{E}_\mu \left[\mathbb{I}_{c>s} \cdot z^{\text{Hor}} \cdot \Delta v_f \right]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s} Z_j^f \left[v_f(\bar{m}, 1, 1) - v_f(m, k, n) \right]}_{\text{Fringe Entry}} \left. \right\}, \tag{27}
\end{aligned}$$

where the challenger superstar entry term is

$$\mathbb{E}_\mu \left[\mathbb{I}_{c>s} \cdot z^{\text{Hor}} \cdot \Delta v_f \right] \equiv \sum_{m_c, k_c, n_c} \mathbb{I} \left\{ \lambda (\theta^{k_c})^\alpha n_c^{1-\alpha} > (\theta^{k_s})^\alpha n_s^{1-\alpha} \right\} z^{\text{Hor}}(m_c, k_c, n_c) \mu(m_c, k_c, n_c) \Delta v_f.$$

with $\Delta v_f \equiv v_f(1, k_s, n_s) - v_f(m, k, n)$ denoting the value change for the fringe firm upon

facing a new incumbent. The first term captures the value gain from radical innovation: the indicator $\mathbb{I}_{f>s'}$ equals one when the fringe firm's innovation is sufficient to overcome the incumbent's total managerial capacity, upon which it enters as a new superstar with state $(\bar{m}, 1, 1)$. The remaining terms account for changes in the competitive environment arising from the incumbent's own innovations, challenger superstar entry, and successful radical innovation by another fringe firm, each of which alters the state the fringe firm faces.

The first-order conditions of the superstar and fringe value functions in equations (26) and (27) yield the following optimal innovation rates along the balanced growth path:

$$z_j^{\text{Hor}} = \left(\frac{\mathbb{P}_{s \geq s'} [v_j((m_j, 1), k, n+1) - v_j(m_j, k, n)]}{\gamma^{\text{Hor}} \cdot \vartheta^{\text{Hor}}} \right)^{\frac{1}{\vartheta^{\text{Hor}} - 1}} \quad (28)$$

$$z_j^{\text{Ver}} = \left(\frac{v_j(m_j + 1, k, n) - v_j(m_j, k, n)}{\gamma^{\text{Ver}} \cdot \vartheta^{\text{Ver}}} \right)^{\frac{1}{\vartheta^{\text{Ver}} - 1}} \quad (29)$$

$$z_j^{\text{Emb}} = \left(\frac{v(\mathbf{m}, k+1, n) - v(\mathbf{m}, k, n)}{\gamma^{\text{Emb}} \cdot \vartheta^{\text{Emb}}} \right)^{\frac{1}{\vartheta^{\text{Emb}} - 1}} \quad (30)$$

$$z_f = \left(\frac{\mathbb{I}_{f>s} [v(\bar{m}, 1, 1) - v_f(m, k, n)]}{\gamma^f \cdot \vartheta^f} \right)^{\frac{1}{\vartheta^f - 1}} \quad (31)$$

Each optimal innovation rate is increasing in the value gain from the corresponding innovation and decreasing in its cost scale and curvature parameters.

The law of motion of the distribution $\mu_t(m, k, n)$ is driven by net flows into and out of state (m, k, n) . Inflows arise from firms that transition into (m, k, n) following a successful innovation along one or more dimensions; outflows arise from firms that leave (m, k, n) as a result of their own innovation or displacement through creative destruction. The first three terms capture inflows from states one step behind in a single dimension; the next three capture inflows from states one step behind in two dimensions; and the subsequent term captures inflows from states one step behind in all three dimensions simultaneously. The final terms capture outflows due to the incumbent's own innovation, displacement

by a fringe firm, and displacement by a challenger superstar.¹³

$$\begin{aligned}
\dot{\mu}_t(m, k, n) = & z_t^{\text{Ver}}(m-1, k, n) \mu_t(m-1, k, n) + z_t^{\text{Emb}}(m, k-1, n) \mu_t(m, k-1, n) \\
& + \mathbb{P}_{s \geq s'} z_t^{\text{Hor}}(m, k, n-1) \mu_t(m, k, n-1) \\
& + z_t^{\text{Ver}}(m-1, k-1, n) z_t^{\text{Emb}}(m-1, k-1, n) \mu_t(m-1, k-1, n) \\
& + z_t^{\text{Ver}}(m-1, k, n-1) \mathbb{P}_{s \geq s'} z_t^{\text{Hor}}(m-1, k, n-1) \mu_t(m-1, k, n-1) \\
& + z_t^{\text{Emb}}(m, k-1, n-1) \mathbb{P}_{s \geq s'} z_t^{\text{Hor}}(m, k-1, n-1) \mu_t(m, k-1, n-1) \\
& + z_t^{\text{Ver}}(m-1, k-1, n-1) z_t^{\text{Emb}}(m-1, k-1, n-1) \\
& \times \mathbb{P}_{s \geq s'} z_t^{\text{Hor}}(m-1, k-1, n-1) \mu_t(m-1, k-1, n-1) \\
& - z_t^{\text{Ver}}(m, k, n) \mu_t(m, k, n) - z_t^{\text{Emb}}(m, k, n) \mu_t(m, k, n) \\
& - \mathbb{P}_{s \geq s'} z_t^{\text{Hor}}(m, k, n) \mu_t(m, k, n) - \mathbb{I}_{f > s} Z_t^f(m, k, n) \mu_t(m, k, n) - Z_{c > s}^{\text{Hor}} \mu_t(m, k, n)
\end{aligned} \tag{32}$$

The labor market clears according to

$$1 = \int_0^1 (l_{sjt} + l_{fjt}) dj. \tag{33}$$

Using equations (24) and (33), the normalized wage satisfies

$$\omega_t = \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(\frac{\phi_t(m, k, n)}{\sigma_t(m, k, n)} + 1 - \phi_t(m, k, n) \right) \mu_t(m, k, n). \tag{34}$$

Combining the production functions in equations (3), (6), and (7) with the labor demand equations (24) yields aggregate output

$$Y_t = Q_t \omega_t^{-1} \exp \left(\int_0^1 \ln \left[\Xi(e_{st}) \left(\frac{((1-\xi)e_{st})^\alpha \phi_{sjt}}{\gamma n_{st}^{\alpha_s} \sigma_{sjt}} \right)^\varepsilon + (1 - \Xi(e_{st})) \left(\lambda^{-m_{jt}} (1 - \phi_{sjt}) \right)^\varepsilon \right]^\frac{1}{\varepsilon} dj \right), \tag{35}$$

where $Q_t = \exp \left(\int_0^1 \ln q_{sjt} dj \right)$ denotes the aggregate quality. Along the balanced growth

¹³At the boundaries of the state space, the relevant outflow terms are set to zero. When any dimension attains its upper bound $(\bar{m}, \bar{k}, \bar{n})$, outflows from further innovation in that dimension are zero.

path, the economy grows at the rate¹⁴

$$g = \ln \lambda \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(z^{\text{Ver}}(m, k, n) + \mathbb{P}_{s \geq s'} z^{\text{Hor}}(m, k, n) + Z^f(m, k, n) \right) \mu(m, k, n). \quad (36)$$

Finally, the resource constraint requires that aggregate output is allocated between consumption and investment across all sectors:

$$Y_t = C_t + \int_0^1 (I_{jt}^{\text{Ver}} + I_{jt}^{\text{Hor}} + I_{jt}^{\text{Emb}}) dj + \int_0^1 I_{jt}^f dj, \quad \text{with} \quad I_{jt}^f = \int_{F_j} I_{ijt} di. \quad (37)$$

Definition (Markov Perfect Equilibrium). A Markov Perfect Equilibrium consists of an allocation $\{C_t, Y_t, y_{sjt}, y_{fjt}\}$, prices $\{r_t, w_t, p_{sjt}, p_{fjt}\}$, innovation policies $\{z^{\text{Ver}}, z^{\text{Hor}}, z^{\text{Emb}}, z_f\}$, labor allocations $\{l_{sjt}, l_{fjt}\}$, and a distribution $\mu_t(m, k, n)$ such that: (i) the final good producer maximizes profit given prices; (ii) the superstar firm maximizes its value function given state $(\mathbf{m}, k, n)$, choosing innovation rates as in equations (28)–(30); (iii) the fringe firm chooses its optimal innovation rate as in equation (31); (iv) the labor market clears; (v) the distribution $\mu_t(m, k, n)$ satisfies equation (14) and evolves according to equation (32); and (vi) aggregate consumption and output grow at the common rate g given by equation (36), with the resource constraint (37) satisfied.

3. Dataset and Empirical Facts

3.1. Data Description

The empirical strategy centers on documenting how markups, productivity, intangible investment composition, and competitive threats vary across the scope distribution. The main dataset is constructed by combining two Compustat sources — Compustat Fundamentals and the Compustat Segment dataset — supplemented by the product market fluidity metric from [Hoberg et al. \(2014\)](#) and the patent-value dataset from [Kogan et al. \(2017\)](#).

¹⁴See Appendix B.5 for details.

Compustat Fundamentals provides comprehensive firm-level financial information for publicly listed companies in North America¹⁵, with extensive longitudinal coverage of balance sheet items, income statement components, cash flow data, and key financial ratios. Measuring firm scope additionally requires information on the industries in which a firm operates. Compustat provides two segment datasets: (i) the Historical Segment dataset¹⁶, which offers long-run coverage of firms' segment activities but limited segment-level detail, and (ii) the Compustat Segment dataset, introduced in 2016, which provides richer segment-level information over a shorter time horizon. In both datasets, a firm's segments are defined as the two-digit industries in which it operates.¹⁷ To balance coverage and detail, I combine the two sources, prioritizing the Compustat Segment dataset when available and relying on the Historical Segment dataset otherwise.¹⁸ I define firm scope as the number of two-digit industries in which a firm operates in a given year, and merge this classification with Compustat Fundamentals to construct the firm-level measures of markups, productivity, and intangible investment composition that form the basis of the empirical analysis.

To capture the competitive threat firms face, I use the product-market fluidity metric from [Hoberg *et al.* \(2014\)](#), which measures how rapidly rivals in similar markets change their product or service offerings. The metric is constructed using natural language processing on product descriptions from firms' annual 10-K filings with the U.S. Securities and Exchange Commission, tracking year-over-year changes in how companies describe

¹⁵Compustat North America includes foreign firms cross-listed in the U.S. via American Depositary Receipts (ADRs), such as Toyota and Unilever. These firms are subject to SEC regulation, comply with U.S. reporting requirements, and actively compete in U.S. product markets, making them economically relevant for the study of firm dynamics. As their exclusion reduces the sample size by approximately 15% without altering the main conclusions, they are retained throughout the analysis. Table [A2](#) provides a comparison of summary statistics.

¹⁶Under Regulation SFAS No. 131 — codified as ASC 280 after 2009 — U.S. public firms are required to disclose the identity of any customer accounting for more than 10% of total revenue, along with the nature of products or services provided. These mandated disclosures constitute the foundation of the Historical Segment dataset.

¹⁷For a comparative illustration of firm segment classification, see Table [A1](#).

¹⁸From the Historical Segment dataset, geographic and operational segments are excluded; only business segments are retained. From the Compustat Segment dataset, only non-missing entries from the Product-Service (PD-SRVC) category are included.

their business activities. A high fluidity score indicates that rivals are rapidly reconfiguring their market positions, intensifying the competitive threat the firm faces.

To trace how firms respond to innovation, I use the patent-value dataset from [Kogan *et al.* \(2017\)](#), which provides patent identifiers, filing and grant dates, firm identifiers, forward citations, and a measure of patent value. The patent value measure is constructed from stock market reactions within a narrow event window around patent grant dates, capturing the surprise component of the market's response to each grant. This short-window design isolates the incremental information content of the grant itself, abstracting from anticipation effects and concurrent firm-level news.

3.2. Measurement

I estimate firm-level total factor productivity following [Olley and Pakes \(1996\)](#), who resolve the simultaneity bias inherent in OLS-based production function estimation by exploiting the monotonicity of investment in unobserved productivity to control for the endogeneity of input choices.¹⁹ To estimate firm-level markups, I follow [De Loecker *et al.* \(2020\)](#) and define markups as the ratio of the output elasticity of the variable input to its expenditure share in total sales, where the output elasticity is recovered from the first-stage production function estimation.

Following [Peters and Taylor \(2017\)](#), I measure transferable intangible investment as total R&D expenditure and embedded intangible investment as 30% of Selling, General, and Administrative (SG&A) expenses net of R&D. SG&A encompasses a broad range of expenditures including advertising, employee compensation, and general operational costs. Once R&D is netted out, the remaining SG&A primarily reflects advertising spending and employee-related costs, most of which are routine operating expenses consumed in the current period. However, a portion accumulates within the firm as lasting intangible capital: advertising builds brand recognition and employee compensation develops organizational knowledge and managerial talent. The 30% fraction captures this accumu-

¹⁹For methodological details, see Appendix [A.2.1](#). Table [A3](#) reports output elasticity estimates under alternative production function estimators, following [Akerberg *et al.* \(2015\)](#) and [Gandhi *et al.* \(2020\)](#).

lating component, with the remainder treated as routine operating costs that generate no lasting intangible value.²⁰

3.3. Empirical Facts

To document how firm-level outcomes respond to innovation, I estimate local projections following Jordà (2005), using the patent-value measure of Kogan *et al.* (2017) as an innovation shock. This measure is well-suited for the present analysis: its market-based construction isolates the unexpected component of each grant, grant timing is largely predetermined relative to the firm’s strategic decisions, and it assigns a dollar value to each patent, capturing the monetary magnitude of each innovation rather than treating all grants as qualitatively equivalent.²¹ The local projection specification takes the form

$$\Delta_h Y_{ijt} = \alpha_h + \beta_{\text{Multi}} V_{it} D_{it} + \beta_{\text{Single}} V_{it} (1 - D_{it}) + \Gamma_h^\top \mathbf{X}_{i,t-1} + \delta_h Y_{i,t-1} + \theta_j + \lambda_t + u_{ijt}, \quad (38)$$

where $\Delta_h Y_{ijt} \equiv Y_{i,t+h} - Y_{it}$ is the h -period change in outcome Y for firm i in industry j , $V_{it} = \sum_{p \in \mathcal{P}_{it}} \text{PatentValue}_{p,t}$ is the firm-year sum of innovation shocks, and $D_{it} = \mathbf{1}\{i \text{ is multi-sector at } t\}$ indicates multi-sector status. The vector $\mathbf{X}_{i,t-1}$ collects lagged controls, $Y_{i,t-1}$ absorbs pre-shock outcome levels, and θ_j and λ_t denote industry and year fixed effects. The coefficients β_{Single} and β_{Multi} trace the impulse responses of single- and multi-sector firms, respectively, to an equivalent innovation shock. Scope status is defined using each firm’s pre-shock classification and held fixed throughout the estimation window, ensuring that firms that subsequently change their reported segments do not mechanically alter the interaction term.

²⁰The two components of embedded intangibles — brand value and organizational capital — could be identified separately, as advertising expenditure is reported as a distinct line item (XAD) in Compustat. However, XAD is sparsely reported in Compustat, and conditioning on non-missing values would disproportionately retain firms with large advertising budgets, introducing a systematic selection bias toward advertising-intensive firms.

²¹Segment changes are infrequent in the data, suggesting that the innovation shocks identified here predominantly reflect vertical improvements within existing sectors rather than scope-expanding horizontal innovations.

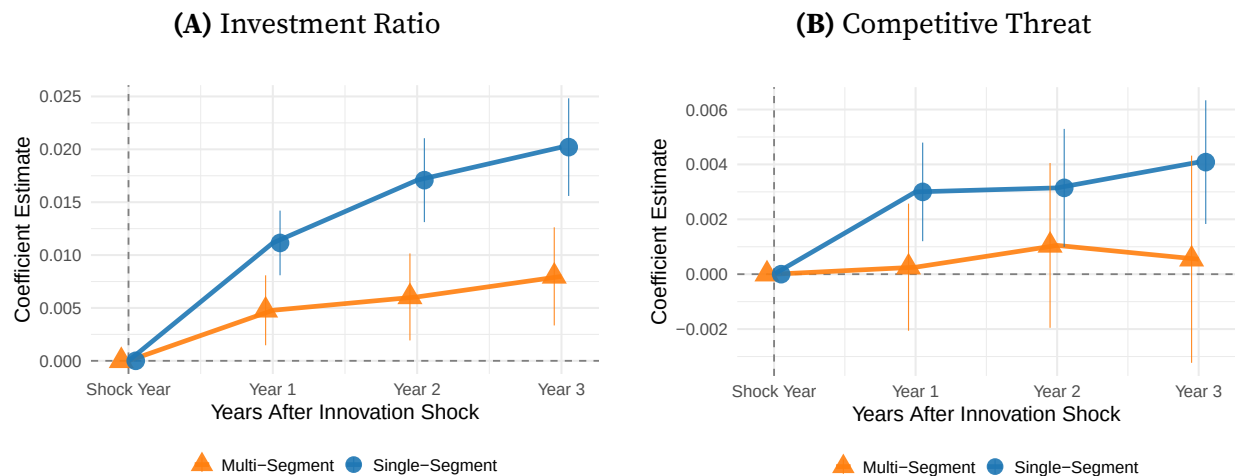


Figure 2. Investment Ratio and Competitive Threat Responses to Innovation Shocks

Note: Note: The innovation shock is measured as $\log(1 + V_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Shaded bands denote 95% confidence intervals based on standard errors clustered at the industry level.

Innovation shocks generate markedly different responses across firm types, and the divergence is consistent across every margin examined. Single-sector firms reallocate investment sharply toward transferable intangibles following a shock, with the response opening at the shock year and widening over the subsequent three years (Figure 2A). Multi-sector firms exhibit a substantially weaker reallocation, suggesting that operating across multiple sectors dilutes the incentive to concentrate innovative investment in a single domain.

The competitive consequences of innovation reinforce this pattern. Single-sector innovators attract significant rival entry following a shock, indicating that their innovations diffuse outward and reshape the competitive landscape, while multi-sector firms experience no discernible increase in competitive threat (Figure 2B). This near-zero response admits two complementary interpretations: multi-sector firms may erect stronger entry barriers through the breadth of their market presence, or their innovations may rest on cross-sector complementarities that narrower rivals cannot easily replicate. Either mechanism implies that multi-sector firms appropriate a larger share of innovation gains internally, insulating themselves from the competitive spillovers that single-sector innovators

face.

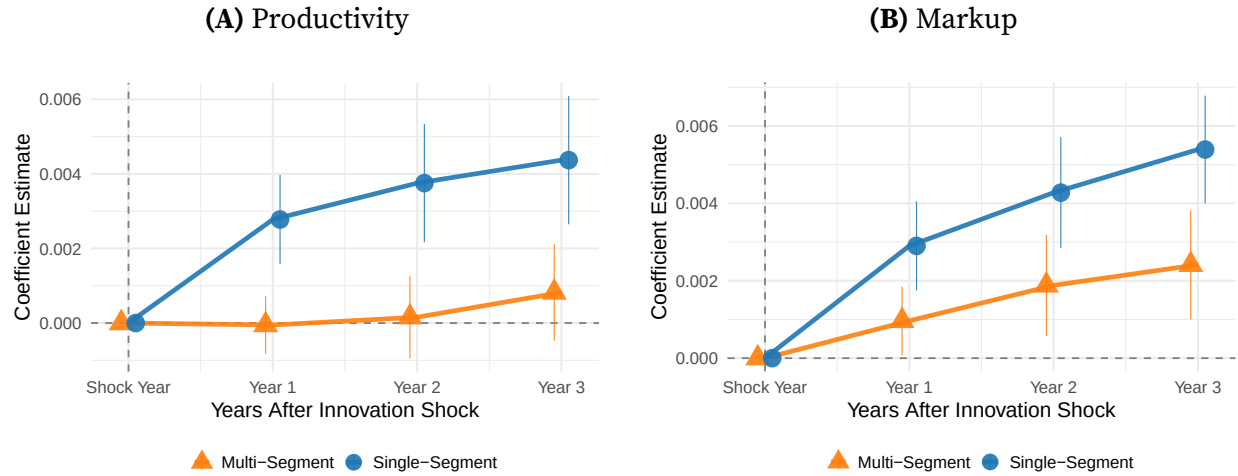


Figure 3. Productivity and Markup Responses to Innovation Shocks

Note: The innovation shock is measured as $\log(1 + V_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Shaded bands denote 95% confidence intervals based on standard errors clustered at the industry level.

The heterogeneity between firm types extends to productivity and markups, the margins that most directly capture the appropriability of innovation rents. Single-sector firms display a strong and monotonically increasing productivity response following an innovation shock, with the coefficient rising steadily from the shock year through year three (Figure 3A). Multi-sector firms, by contrast, exhibit a response that remains flat and statistically indistinguishable from zero throughout the entire horizon, with the gap between the two firm types widening persistently over time.

A parallel pattern emerges for markups, where the scope gap is present but more moderate (Figure 3B). Single-sector firms command significantly higher price-cost margins following an innovation shock, with the markup response growing monotonically and precisely estimated at all horizons. Multi-sector firms also exhibit a positive and increasing markup response, but the magnitude is substantially smaller throughout the horizon, and the gap between the two firm types widens persistently from year one onward. Taken together, both panels point to the same conclusion: firm scope systematically attenuates the returns to innovation. Single-sector firms translate innovation shocks into lasting

gains in productive efficiency and market power, while multi-sector firms appropriate substantially smaller rents from equivalent shocks. This finding is consistent with [Autor *et al.* \(2020\)](#), in which higher-productivity firms command higher markups, and extends it by showing that scope moderates the productivity-markup nexus.²²

Taken together, these results point to a coherent set of regularities, summarized by the following three stylized facts.²³

Fact 1. Firm scope attenuates the transferable intangible investment response to innovation shocks: single-sector firms exhibit a stronger and more persistent reallocation toward transferable intangibles than their multi-sector counterparts

Fact 2. Innovation diffuses outward in focused firms. Following a shock, single-sector firms face a measurable increase in competitive threat from rivals, while multi-sector firms exhibit no significant competitive response at any horizon.

Fact 3. The gains from innovation in both productivity and markups are concentrated in single-sector firms. Multi-sector firms display significantly weaker responses along both margins, indicating that scope limits the appropriability of innovation rents.

4. Quantitative Analysis

4.1. Calibration and Identification

For scope level n , the marginal distribution is

$$\mu(n) = \sum_m \sum_k \mu(m, k, n).$$

The competitive threat at scope level n is defined as the probability that an incumbent loses a given sector, conditional on a challenger's successful innovation, combining dis-

²²Appendix Figure A1 replicates the analysis using productivity and markup estimates based on [Akerberg *et al.* \(2015\)](#) and [Gandhi *et al.* \(2020\)](#)

²³Appendix Table A4 confirms that these patterns hold under a two-way fixed effects specification, providing a complementary check on the baseline local projection results.

placement by a rival superstar and by a fringe firm:

$$CT(n) = \mathbb{E}_n[\mathbb{P}_{c>s'}(m, k, n) + \mathbb{I}_{f>s'}(k, n)],$$

Investment ratio and the average markup at scope level n are

$$I(n) = \mathbb{E}_n \left[\frac{I^{\text{Ver}}(m, k, n) + I^{\text{Hor}}(m, k, n) + I^f(m, k, n)}{I^{\text{Emb}}(m, k, n)} \right], \quad \sigma(n) = \mathbb{E}_n \left[\frac{1 - \varepsilon \phi(m, k, n)}{(1 - \phi(m, k, n)) \varepsilon} \right].$$

Finally, the growth rate and productivity²⁴ at scope level n are

$$g(n) = \mathbb{E}_n \left[\ln \lambda \left(z^{\text{Ver}}(m, k, n) + \mathbb{P}_{k \geq k'} z^{\text{Hor}}(m, k, n) + Z^f(m, k, n) \right) \right], \quad Q(n) = e^{g(n)}.$$

The model is disciplined by 36 empirical moments, comprising 18 targeted and 18 untargeted moments across six scope levels. The calibration relies on 16 parameters, of which four are set externally. The time discount rate is fixed at $\rho = 0.05$, and the curvature parameters governing R&D investment for vertical innovation, horizontal innovation, and fringe firms are set to $\vartheta^{\text{Ver}} = \vartheta^{\text{Hor}} = \vartheta^f = 2.0$, following [Akcigit and Kerr \(2018\)](#). The remaining 12 parameters ($\varepsilon, \lambda, \theta, \alpha, \gamma, \beta, \xi, \gamma^{\text{Ver}}, \gamma^{\text{Hor}}, \gamma^f, \gamma^{\text{Emb}}, \vartheta^{\text{Emb}}$) are estimated internally and pin down the model's key margins. Table 1 reports all parameter values. Appendix ?? details the construction of empirical counterparts, while Appendix C.2 describes the solution method.

4.2. Model Performance

Figure 4 shows that the model matches the targeted moments closely across all dimensions. Panel (A) reproduces the monotone decline in markups with scope, capturing both its slope and magnitude; the remaining discrepancies are confined to the extremes. Panel (B) shows that the model tightly tracks the transferable-to-embedded investment ra-

²⁴Productivity satisfies $\dot{Q}(n, t)/Q(n, t) = g(n)$, which integrates to $Q(n, t) = Q(n, t_0) e^{g(n)(t-t_0)}$ for any normalization date t_0 . Hence, $Q(n)$ denotes the associated gross growth factor over a unit interval. The choice of normalization affects only levels and has no bearing on equilibrium allocations or identification.

Table 1. Parameter Values

Parameter	Description	Value
-----External Calibration-----		
ρ	Discount rate	0.050
ϑ^{Ver}	Curvature of vertical investment	2.000
ϑ^{Hor}	Curvature of horizontal investment	2.000
ϑ^f	Curvature of fringe firms investment	2.000
-----Internal Calibration-----		
ε	CES parameter	0.864
λ	Quality improvement step size	1.084
θ	Embedded innovation step size	1.148
α	Curvature of managerial quality	0.591
γ	Scale of managerial quality	0.759
β	Curvature of brand value	0.108
ξ	Share of brand value	0.347
γ^{Ver}	Cost scale of internal innovation	4.188
γ^{Hor}	Cost scale of horizontal innovation	4.022
γ^f	Cost scale of fringe	194.160
γ^{Emb}	Cost scale of embedded innovation	285.883
ϑ^{Emb}	Curvature of embedded innovation	5.577

Note: The upper limit for the number of production lines $\bar{n}$ is set to 6, and the upper bounds for $\bar{m}$ and $\bar{k}$ are set to 9.

tio, including the sharp drop from $n = 1$ to $n = 2$ and the subsequent flattening. Panel (C) demonstrates that the model replicates the scope distribution almost exactly, capturing both the concentration of superstar firms at narrow scope and the rapid thinning as scope expands. Overall, the model fits the targeted moments with high accuracy.

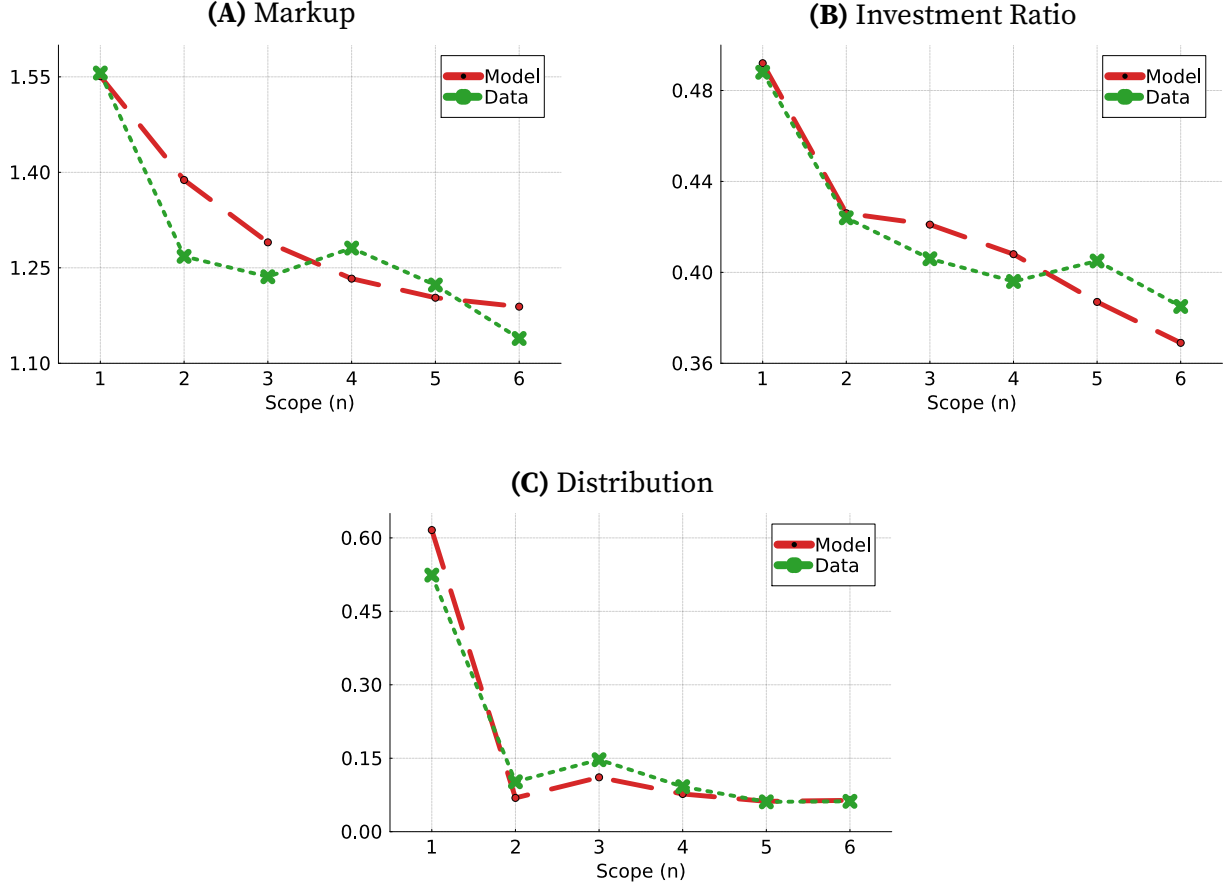


Figure 4. Targeted Moments: Markup, Investment Ratio and Distribution by Scope

Note: The green line represents the data and the red line the model along the balanced growth path. The sample excludes firms in the utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A expenditures. Markups, the investment ratio, and the size distribution are measured for the 2019 cross-section. The investment ratio is defined as transferable over embedded investment (see Section 3.2) and is winsorized at the 95th percentile; markups are winsorized at the 90th percentile. Appendix ?? provides details on the construction of empirical counterparts to the model moments.

Figure 5 evaluates the model along untargeted dimensions. The model continues to perform strongly, indicating that the mechanisms disciplined by the targeted moments generalize. Panel (A) shows that the model reproduces the decline in productivity with scope with only minor deviations. Panel (B) captures the overall downward trend in growth rates, although it does not reproduce the non-monotonicities at intermediate scope levels; in particular, it understates growth for narrow-scope firms and does not generate negative growth at $n = 6$. Panel (C) shows that the model closely matches the competitive threat profile, including its sharp decline from $n = 1$ to near zero at high scope. Taken

together, the model delivers a tight fit to moments that were not targeted in estimation, providing strong support for its quantitative discipline and for the structural interpretation of the underlying parameters.

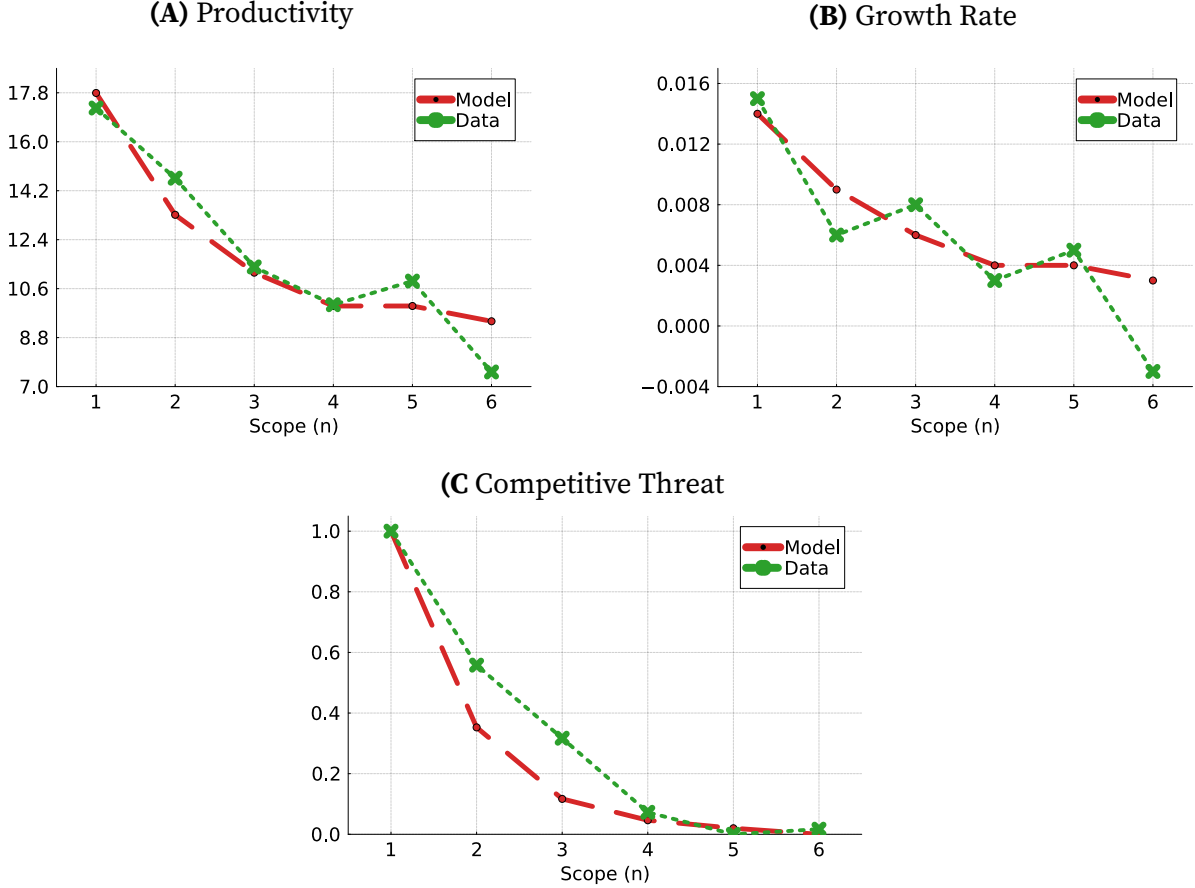


Figure 5. Untargeted Moments: Productivity, Growth Rate and Competitive Threat by Scope

Note: The green line represents the data and the red line the model along the balanced growth path. The sample excludes firms in the utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A expenditures. Productivity and competitive threat are measured for the 2019 cross-section and are described in Section 3.2. The growth rate is defined as the two-year log change in productivity, $\Delta_2 \ln(\text{prod})_{i,t} = \ln(\text{prod}_{i,t}) - \ln(\text{prod}_{i,t-2})$, averaged over 2005–2019 and winsorized at the 10th and 90th percentiles and productivity is winsorized at the 95th percentile. Competitive threat, in both the model and data, is computed as the average at each scope level and rescaled via min–max normalization, $(\bar{x}(n) - \min \bar{x}(n)) / (\max \bar{x}(n) - \min \bar{x}(n))$, which lies in $[0, 1]$, with higher values indicating greater competitive pressure. Appendix ?? details the construction of empirical counterparts to the model moments.

4.3. Superstar Firms Innovation Direction and Incentives

Figure 6 illustrates how the optimal innovation direction of superstar firms varies across scope, the transferable intangible gap m , and the embedded intangible level k . Two broad patterns emerge. First, scope itself exerts a first-order effect on innovation incentives: as

firms expand from $n = 1$ to $n = 6$, the span-of-control constraint progressively erodes market share in each individual sector. Second, within each scope level, substantial heterogeneity persists depending on the composition of a firm's intangible endowment, and this relationship is highly nonlinear, particularly at low scope.

At $n = 1$, the specialization index reveals a rich interaction between the two intangible dimensions. Firms with low accumulated embedded intangibles optimally pursue horizontal innovation: without sufficient embedded stocks, cross-sector complementarities are weak, and expanding scope offers higher returns than deepening quality within a single product line. As k increases, however, embedded intangibles amplify the complementarity with transferable intangibles and raise the firm's market share in its current sector, tilting incentives toward vertical innovation. Yet this pattern is moderated by the transferable intangible gap: as m rises, the return to vertical quality improvement declines, and sufficiently high embedded intangible levels further raise the return to scope expansion—since embedded intangibles are freely mobile across sectors—shifting firms back toward horizontal innovation. In the boundary case $m = \bar{m}$, firms at the maximum transferable gap can no longer improve quality vertically, making these regions absorbing states that concentrate innovation incentives entirely toward the horizontal margin, as reflected in the deep red cells along the rightmost column.

At $n = 2$, a similar logic applies, but the span-of-control constraint begins to bind more tightly. Firms with sufficiently high embedded intangible levels no longer find it profitable to enter an additional sector: the market share loss induced by the span-of-control constraint across existing lines outweighs the gains from scope expansion, and vertical innovation dominates. Firms with lower embedded levels, whose market share in each sector is more modest, retain stronger incentives to invest horizontally, though these incentives are already attenuated relative to $n = 1$. This attenuation grows monotonically with scope: the horizontal innovation region shrinks from $n = 2$ onward, and by $n = 4$ it nearly vanishes entirely. A stark illustration arises at $n = 3$ with $m = \bar{m}$: firms at the maximum transferable gap cannot improve quality vertically, yet for embedded levels above $k = 2$ they also refrain from horizontal investment, since the market share dilution

implied by the span-of-control constraint exceeds the gains from entering an additional sector.

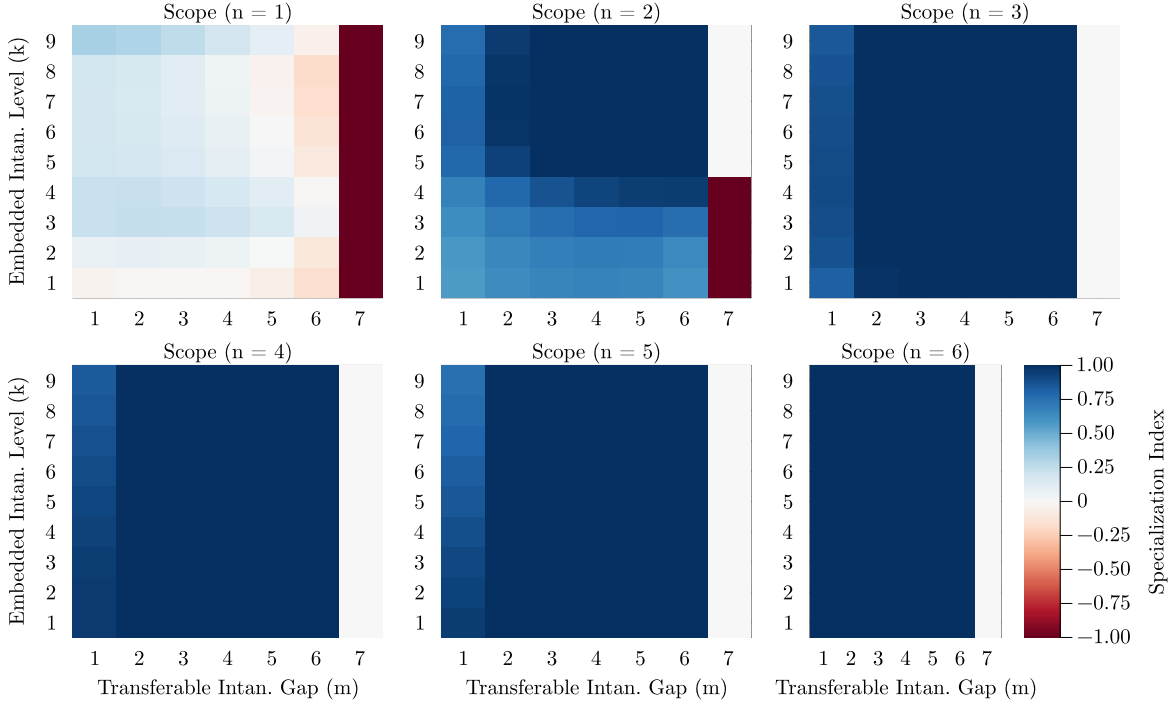


Figure 6. Superstar Firms Innovation Direction (Vertical vs Horizontal)

Each panel displays the specialization index $S(m, k) = \frac{z^{Ver}(m, k) - \mathbb{P}_{s > s^*} z^{Hor}(m, k)}{z^{Ver}(m, k) + \mathbb{P}_{s > s^*} z^{Hor}(m, k)}$ evaluated over the (m, k) grid for a given scope level n . The index lies in $[-1, 1]$: blue cells indicate specialization toward vertical innovation, red cells indicate specialization toward horizontal innovation, and white cells indicate approximate indifference between the two directions. Each panel holds scope fixed and varies the transferable intangible gap m along the horizontal axis and the embedded intangible level k along the vertical axis, with the color gradient within a panel capturing how the innovation direction of superstar firms responds to their intangible stock composition at a given scale. Panels are arranged in order of increasing scope from $n = 1$ (top left) to $n = 6$ (bottom right).

A complementary pattern emerges at the boundary $m = \bar{m}$ as scope increases. At low scope ($n = 1, 2$), the maximum transferable gap is an absorbing state that concentrates incentives sharply toward horizontal innovation, as reflected in the deep red cells along the rightmost column. This character changes markedly as scope expands: by $n \geq 3$, the rightmost column transitions to white, indicating near-indifference between the two innovation directions. The economic mechanism is distinct from the interior blue region. In the interior, the span-of-control constraint drives firms toward vertical investment by making scope expansion prohibitively costly. At the boundary $m = \bar{m}$, however, firms cannot improve quality vertically, yet at high scope the market share dilution from horizontal entry is also too severe to justify further expansion. The result is approximate

indifference—not a preference for vertical innovation, but an exhaustion of profitable margins in both directions.

Taken together, Figure 6 establishes that innovation direction cannot be reduced to firm size alone. Scope shapes the overall level of innovation incentives by governing market share and managerial costs, while the composition of the intangible endowment determines the direction of investment within each scope level. The interaction between these two dimensions is inherently nonlinear—manifesting in the sharp interior heterogeneity at low scope, the near-universal dominance of vertical innovation at high scope, and the qualitative shift in boundary behavior at $m = \bar{m}$ as scope expands—and both are necessary to account for the heterogeneity in innovation strategies observed across superstar firms.

5. Counterfactual Analysis

5.1. Shutting Down Embedded Intangibles and Span of Control Constraint

To understand the effects of the span-of-control constraint and embedded intangibles on scope distribution, long-run economic growth, markups, and competitive threat, I consider two counterfactuals: shutting down the span-of-control constraint ($n^\alpha = 1$) and eliminating embedded intangibles ($\bar{k} = 1$).

Shutting down the span-of-control constraint allows superstar firms to expand scope without decreasing managerial quality per sector; consequently, the scope distribution shifts mass toward higher scope levels, since horizontal innovation no longer erodes managerial quality. Markups rise monotonically with scope, reversing the baseline pattern. The growth rate, in turn, exhibits an inverted-U pattern with scope.

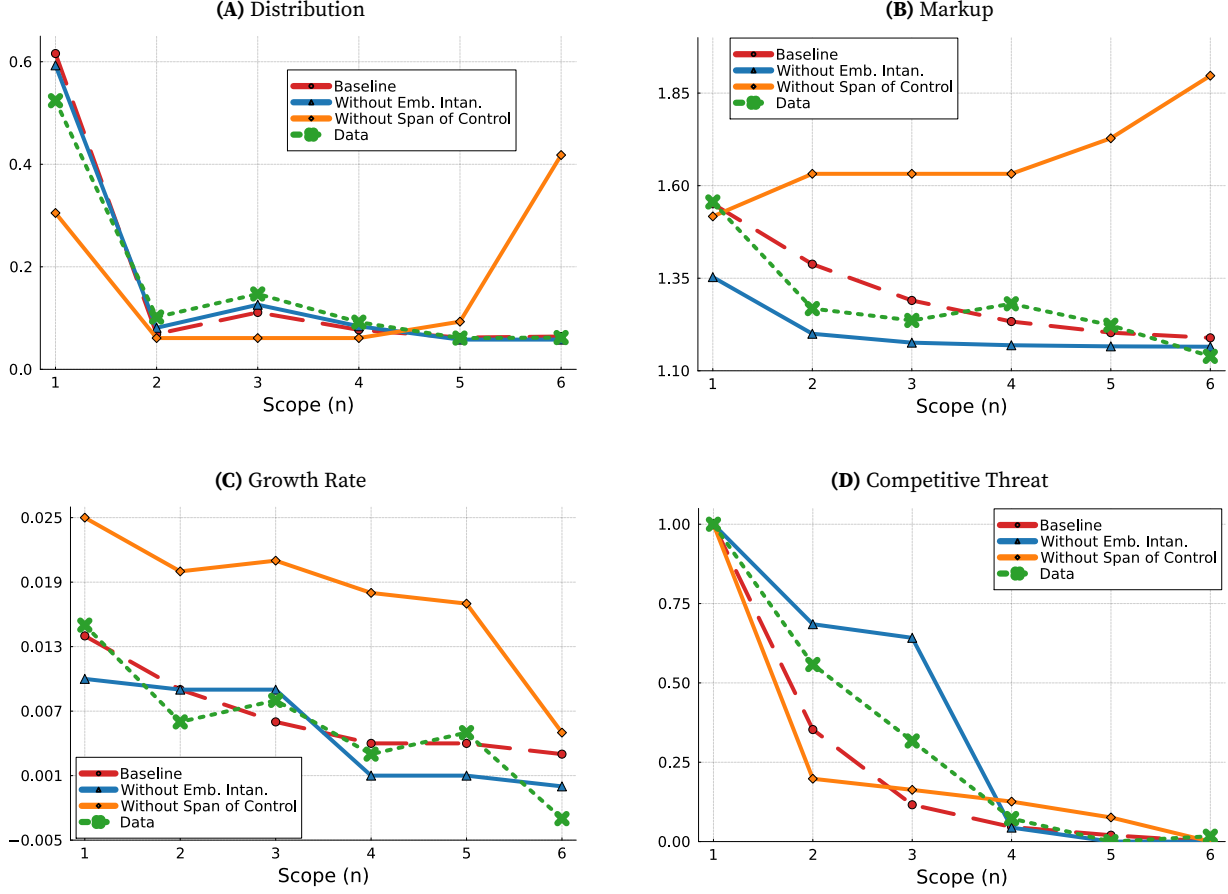


Figure 7. Counterfactual Analysis Under Different Cases

Note: The green line represents the data, whereas the red line shows the model results along the balanced growth path. The orange line corresponds to the case with the span-of-control channel shut down ($n^\alpha = 1$), while the blue line corresponds to the case with embedded intangibles shut down ($k = 1$). Appendix ?? provides details on the calculation of the model moments. The reported competitive threat measure is computed as the average at each scope level and then rescaled using a min-max normalization, $(x - \min x) / (\max x - \min x)$, to lie in $[0, 1]$, capturing relative variation across scope levels, with higher values indicating greater competitive pressure.

At narrow scope, incumbent firms invest more intensively in innovation, driving growth above the baseline. As scope broadens, however, these same firms erect progressively higher intangible-based entry barriers, compressing competitive threat and attenuating innovation incentives at the frontier. Growth therefore peaks at intermediate scope levels and declines thereafter, though it remains above the baseline throughout—the sole exception being single-product firms, where growth falls below both the baseline and the data.

When embedded intangibles are eliminated, the scope distribution is largely unchanged, and the markup profile is nearly flat across scope levels. The sharpest departure from the data arises along the competitive threat margin: without intangible-based entry barriers,

competitive threat diverges markedly from both the baseline and the empirical pattern.

The two counterfactuals reveal distinct and complementary roles: the span-of-control constraint disciplines scope expansion, generates the declining markup gradient, and shapes the inverted-U growth profile across scope levels, while embedded intangibles govern the competitive threat profile and underwrite aggregate innovation. Neither channel alone is sufficient to account for the full set of empirical regularities documented in Section 3.3.

5.2. Shutting Down Entry Barriers and Misallocation

Misallocation in the model operates through two distinct channels: markup dispersion across firms, and the joint entry barrier of scope and embedded intangibles that prevents potential entrants from capturing a market unless their total managerial capacity exceeds the incumbent's. To quantify the first channel, I adapt Peters (2020) and decompose aggregate output as $Y_t = Q_t \times E_t \times M_t \times S_t$, where Q captures quality improvements, M captures misallocation from markup dispersion, and E and S are residual terms.²⁵ The misallocation term M is defined as

$$M_t = \frac{\exp \left(\sum_m \sum_k \sum_n \ln \left[\frac{\phi(m, k, n)}{\sigma(m, k, n)} + (1 - \phi(m, k, n)) \right] \mu(m, k, n) \right)}{\sum_m \sum_k \sum_n \left(\frac{\phi(m, k, n)}{\sigma(m, k, n)} + (1 - \phi(m, k, n)) \right) \mu(m, k, n)},$$

namely the ratio of the geometric to the arithmetic mean of firm-level output shares, which endogenously evolves with markup dispersion.²⁶ To analyze the second channel, I conduct three counterfactual experiments that progressively dismantle the entry barriers

²⁵For details see Appendix B.6.

²⁶The term M differs slightly from Peters (2020): even with constant markups across superstar firms, dispersion between superstars and fringe firms persists through the market share ϕ .

embedded in the creative destruction conditions of the model.²⁷

The first removes only the scope component of the entry barrier, isolating the role of embedded intangibles in protecting incumbents. The second removes only the embedded intangible component, isolating the role of scope. The third eliminates both simultaneously, providing an upper bound on the allocative gains from barrier removal.

Table 2 presents the results. The baseline $M = 0.985$ confirms that markup dispersion alone generates only a modest output loss of around 1.5 percent, placing the primary source of misallocation in entry barriers rather than pricing distortions. Removing the scope barrier marginally improves M to 0.991 but lowers the growth rate from 0.012 to 0.011, suggesting that scope barriers, though mildly distortionary, discipline vertical innovation incentives by preserving the incumbent's managerial capacity advantage across existing production lines. The dominant finding is that eliminating the embedded intangible barrier raises the growth rate to 0.020 while improving M to 0.988, identifying the embedded dimension as the quantitatively decisive source of misallocation.

The embedded barrier is particularly binding for narrow-scope firms: an entrant competing against a single-sector superstar must match its entire embedded stock, whereas the scope barrier only constrains challengers at later stages of horizontal expansion, precisely when the span-of-control constraint has already attenuated innovation incentives independently. The two barriers therefore do not operate symmetrically, as the embedded barrier bites early when incentives are strongest, while the scope barrier bites late

²⁷*Counterfactual 1 (No Scope Barrier).* The scope component $n^{1-\alpha}$ is shut down by setting $n_s = n_{s'} = 1$ in both conditions. The displacement conditions for the challenger superstar and fringe reduce to

$$\lambda (\theta^{k_s})^\alpha > (\theta^{k_{s'}})^\alpha \quad \text{and} \quad \lambda^{\bar{m}} > (\theta^{k_{s'}})^\alpha.$$

Counterfactual 2 (No Embedded Barrier). The embedded intangible component $(\theta^k)^\alpha$ is shut down by setting $k_s = k_{s'} = 1$ in both conditions,

$$\lambda (n_s)^{1-\alpha} > (n_{s'})^{1-\alpha} \quad \text{and} \quad \lambda^{\bar{m}} > (n_{s'})^{1-\alpha}.$$

Counterfactual 3 (No Barriers). Both components are shut down simultaneously, so that any successful horizontal or fringe firm innovations immediately become the new incumbent regardless of the incumbent's embedded stock or scope:

$$\lambda > 1 \quad \text{and} \quad \lambda^{\bar{m}} > 1.$$

when they are already weak.²⁸ Figure C2 in the Appendix confirms this mechanism: removing the embedded barrier substantially raises both transferable intangible investment and horizontal innovation rates among single-scope.

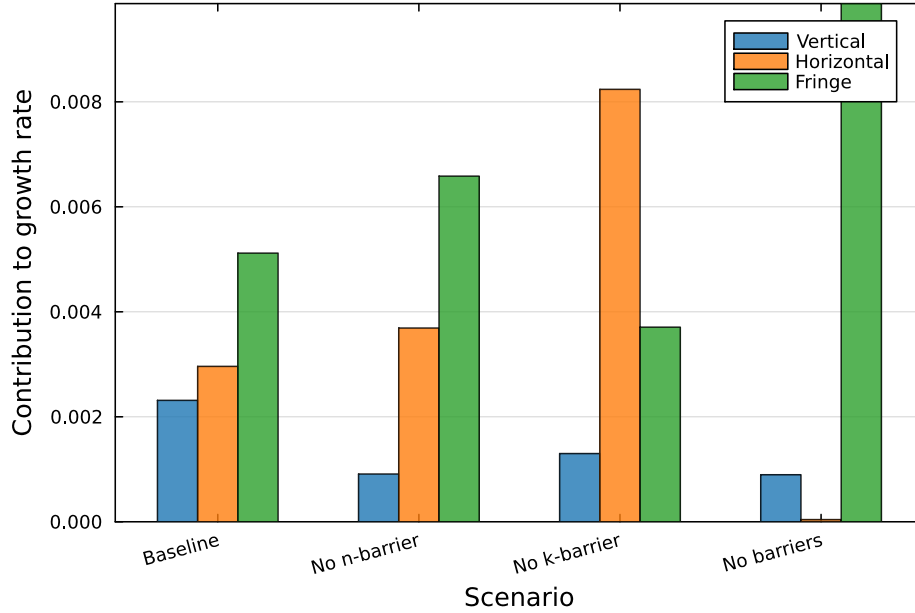


Figure 8. Growth Decomposition by Innovation Type and Entry Barrier

Note: Each line reports the percentage change in consumption equivalence welfare $\Delta\Omega(\%)$ relative to the baseline under alternative policy designs. Three policy instruments are considered: a scope tax τ_{scope} , a horizontal investment tax τ_{hor} , and an embedded intangible investment tax τ_{emb} , each applied in isolation (dashed lines) or combined with a flat subsidy to fringe firms τ_f (solid lines). The horizontal axis indexes the progressivity threshold τ : at the left endpoint (Flat Tax), all firms face a uniform tax regardless of scope; as τ increases, the tax applies only to firms with scope above the threshold, exempting smaller firms; at the right endpoint (Flat Sub.). Detailed thresholds and corresponding numerical results are reported in Appendix C1

Removing all barriers simultaneously achieves the highest $M = 0.993$ but reduces the growth rate to 0.010, as frictionless entry erodes the innovation rents sustaining the vertical quality ladder. This tradeoff reveals that entry barriers, while distortionary from an allocative standpoint, serve as a necessary incentive mechanism: their complete removal improves the static distribution of output across firms but undermines the dynamic forces driving long-run productivity growth through creative destruction.

²⁸Figure C3 in the Appendix reports entry probabilities along the scope n and embedded intangible k dimensions for both challenger superstar and fringe firms.

Table 2. Misallocation and Growth Rate Under Different Entry Barrier Scenarios

	Baseline	No Scope Barrier (n)	No Embedded Barrier (k)	No Barriers
M	0.991	0.995	0.995	0.996
g	0.010	0.011	0.013	0.011

6. Policy Implications

The counterfactual analysis in Section 5.2 establishes that the embedded intangible barrier is the quantitatively dominant source of misallocation, binding most severely for narrow-scope superstar firms where innovation incentives are strongest. Directly taxing embedded intangibles or subsidizing narrow-scope superstar firms is not straightforward, however, since embedded intangibles are complementary to both vertical and horizontal innovation: any instrument that reduces their accumulation also attenuates the innovation incentives they underwrite. I therefore evaluate three policy instruments: a scope-dependent profit tax τ_{scope} , a horizontal investment tax τ_{hor} , and an embedded intangible investment tax τ_{emb} , each applied in isolation and combined with a flat subsidy to fringe entrants τ_f .

Following [Berlingieri et al. \(2025\)](#), I adopt a threshold approach and extend it to incorporate both taxes and subsidies. For the scope-dependent instruments (τ_{scope} and τ_{hor}), the policy is indexed by a scope threshold j : superstar firms with scope $n \leq j$ receive a subsidy of +0.2, while superstar firms with scope $n > j$ face a tax of -0.2 . For the embedded intangible instrument (τ_{emb}), the threshold is instead defined over the embedded intangible level k : superstar firms with $k \leq j$ receive a subsidy of +0.2, while superstar firms with $k > j$ face a tax of -0.2 . In both cases, the design spans the full range from a uniform tax to a uniform subsidy, with intermediate thresholds implementing progressively targeted redistribution.²⁹ Welfare effects are evaluated using a consumption-equivalence measure that quantifies the permanent percentage change in consumption leaving households in-

²⁹Net tax revenue, defined as the difference between taxes collected and subsidies disbursed, is redistributed to households through lump-sum transfers when positive, and financed by lump-sum levies on households when negative.

different between the baseline and the counterfactual.³⁰

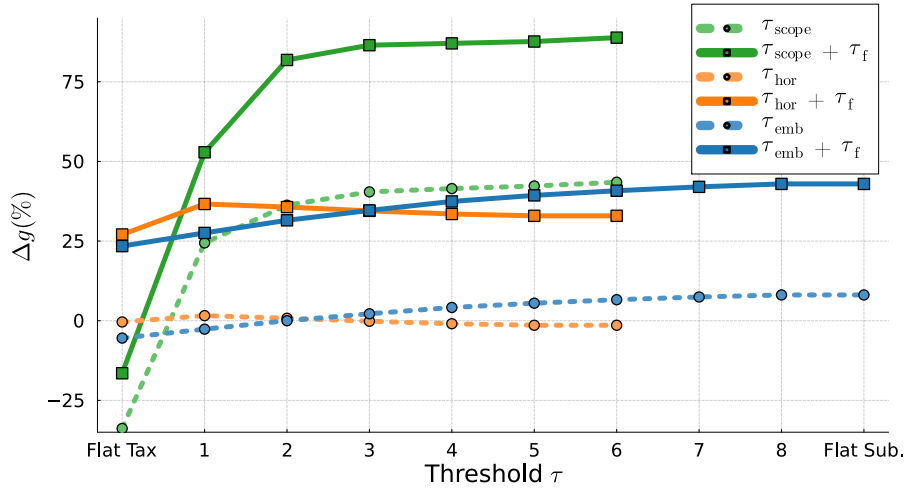


Figure 9. Flat and Progressive Policies Effect on Growth Rate

Note: Each line reports the percentage change in the aggregate growth rate $\Delta g(\%)$ relative to the baseline under alternative policy designs. Three policy instruments are considered: a scope tax τ_{scope} , a horizontal investment tax τ_{hor} , and an embedded intangible investment tax τ_{emb} , each applied in isolation (dashed lines) or combined with a flat subsidy to fringe firms τ_f (solid lines). The horizontal axis indexes the progressivity threshold τ : at the left endpoint (Flat Tax), all firms face a uniform tax regardless of scope; as τ increases, the tax applies only to firms with scope above the threshold, exempting smaller firms; at the right endpoint (Flat Sub.). Detailed thresholds and corresponding numerical results are reported in Appendix C1

Figures 9 and 10 reveal a striking disconnect between growth and welfare across policy designs. The scope tax combined with the fringe subsidy delivers the largest aggregate growth gains at intermediate thresholds, exceeding 50% above baseline, yet the highest welfare gain is achieved by a different design: setting the threshold at $j = 1$, so that single-sector superstar firms receive a subsidy while all superstar firms with $n > 1$ face a tax, combined with the fringe subsidy τ_f . This policy generates a welfare gain of approximately 20%, substantially exceeding all other configurations.

The intuition is precise. Subsidizing single-sector superstar firms operates across all three innovation margins simultaneously: it strengthens vertical quality improvement, encourages horizontal scope expansion, and raises the return to embedded intangible accumulation, since all three are complementary at narrow scope and this is exactly where embedded intangible barriers bind most severely. Policies that instead generate the highest growth rates, in particular flat scope taxes that push superstar incumbents

³⁰See Appendix B.7 for details.

toward vertical innovation, require large upfront investment costs that erode household consumption in the short run. The progressive design with $j = 1$ avoids this tradeoff by encouraging broad-based investment at low cost while taxing wide-scope superstar incumbents, collecting revenue redistributed to households and sustaining consumption alongside growth.

The fringe subsidy τ_f is a necessary complement to any of the three instruments: without it, taxing superstar incumbents merely transfers rents without improving the efficiency of creative destruction, generating welfare losses or negligible gains. This reflects the role of fringe firms as the fastest-growing firms in the economy, since restoring their competitive margin against superstar incumbents generates disproportionate aggregate gains that no incumbent-facing instrument alone can replicate. Among instruments combined with τ_f , the embedded intangible and horizontal investment taxes ($\tau_{\text{emb}} + \tau_f$ and $\tau_{\text{hor}} + \tau_f$) deliver stable positive welfare gains across all thresholds but remain uniformly below the progressive scope tax at $j = 1$. This ranking reflects the targeting advantage of the scope design: conditioning the exemption on firm scope directly addresses the misallocation margin identified in Section 5.2, whereas profit taxes correct intangible accumulation distortions without providing the expansion incentive that single-sector superstar firms require to overcome entry barriers.

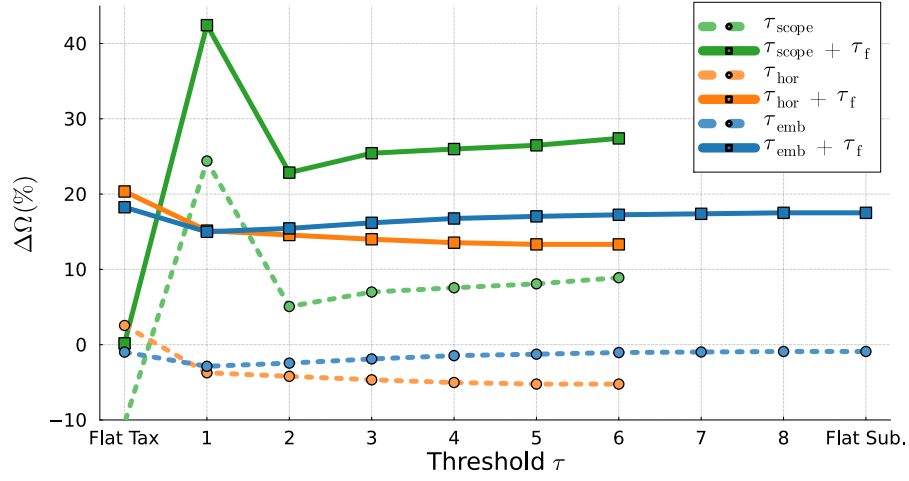


Figure 10. Flat and Progressive Policies Effect on Welfare

Note: Each line reports the percentage change in consumption equivalence welfare $\Delta\Omega(\%)$ relative to the baseline under alternative policy designs. Three policy instruments are considered: a scope tax τ_{scope} , a horizontal investment tax τ_{hor} , and an embedded intangible investment tax τ_{emb} , each applied in isolation (dashed lines) or combined with a flat subsidy to fringe firms τ_f (solid lines). The horizontal axis indexes the progressivity threshold τ : at the left endpoint (Flat Tax), all firms face a uniform tax regardless of scope; as τ increases, the tax applies only to firms with scope above the threshold, exempting smaller firms; at the right endpoint (Flat Sub.). Detailed thresholds and corresponding numerical results are reported in Appendix C1

7. Conclusion

This paper establishes a unified endogenous growth model in which the direction of innovative effort emerges endogenously from firms' expansion decisions. Horizontal expansion introduces span-of-control frictions that reduce managerial quality per sector and lower per-sector profitability. In response, multi-sector firms systematically reallocate investment away from transferable intangibles, which generate knowledge spillovers and fuel creative destruction, and toward embedded intangibles, which provide firm-specific competitive advantages. This reallocation is self-reinforcing: broader scope raises the return to embedded intangibles through economies of scope, inducing further investment in brand value and organizational capital, which in turn lowers the cost of entering additional sectors. As scope and embedded intangible stock grow jointly, the incumbent superstar's total managerial capacity increases, raising entry barriers and suppressing creative destruction — generating a wedge between private and social returns that instruments conditioned on firm size alone cannot correct.

This framework suggests several directions for future research. First, incorporating the evolution of firm scope over the lifecycle would permit a richer analysis of dynamic market segmentation strategies and their long-run consequences for the direction and pace of innovation. Second, a formal comparison between the decentralized equilibrium and a social planner's solution would allow for the precise derivation of optimal policy instruments to address the spillover frictions generated by diversification. Third, the model generates testable predictions regarding how firms respond endogenously to span-of-control constraints — micro-econometric work could examine the relative efficacy of organizational redesign, investment in information technology, and human capital accumulation as adaptive strategies. Finally, a critical open question is whether embedded intangible capital slows idea diffusion by limiting knowledge spillovers. Addressing this with richer microdata and dynamic structural methods will be essential for designing policies that balance the private gains from diversification against the broader social returns to innovation and competition.

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Appendices

A. Empirical Appendix

A.1. Dataset

Table A1. Example Firms Segment in Compustat Segment Dataset

Company	Segments
TOYOTA MOTOR CORP	Financial Services
	Automotive
	All Other
PROCTER & GAMBLE CO	Health Care
	Grooming
	Corporate
	Beauty
	Baby, Feminine & Family Care
TESLA INC	Fabric & Home Care
	Energy Generation & Storage
	Automotive

A.2. Measurement Details

A.2.1. Production Function Estimation

Sample and specification. The estimation sample covers 1990–2019 and excludes the finance and utilities sectors. All estimations are conducted separately for each two-digit NAICS industry. The gross-output production function is specified as

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \epsilon_{it}, \quad (39)$$

where y_{it} is log gross output (sale), l_{it} is log labor (emp), k_{it} is log capital (ppent), and m_{it} is log intermediate inputs (cogs). The term ω_{it} denotes the unobserved firm-level productivity shock observed by the firm when making input decisions, and ϵ_{it} is an iid

measurement error. The sample is restricted to observations satisfying

$$\text{sale} > 0, \quad \text{ppent} > 0, \quad \text{cogs} > 0, \quad \text{emp} > 0, \quad \text{capx} > 0, \quad \text{cogs} < \text{sale}, \quad (40)$$

where the final restriction ensures that intermediate input expenditure does not exceed total revenue.

All nominal variables are deflated to real terms prior to estimation. Output (sale) is deflated using the GDP deflator.³¹ Intermediate inputs (cogs) and capital expenditure (capx) are deflated using the Producer Price Index.³² The capital stock (ppent) is deflated using the Gross Private Domestic Investment deflator.³³

Baseline: Olley and Pakes (1996) (OP). The baseline production function estimates follow Olley and Pakes (1996), who resolve the simultaneity bias inherent in OLS estimation by exploiting the monotonicity of investment in unobserved productivity. Under the assumption that capital expenditure (capx) is strictly increasing in ω_{it} conditional on k_{it} , the productivity shock can be recovered as $\omega_{it} = h(k_{it}, \text{capx}_{it})$, which is approximated nonparametrically in the first stage. The OP estimator is adopted as the baseline because De Loecker *et al.* (2020) recover the output elasticities required for markup identification from the same production function, ensuring internal consistency between the TFP and markup estimates.

Robustness: Akerberg, Caves, and Frazer (2015) (ACF). The first robustness check follows Akerberg *et al.* (2015), who address the collinearity problem in the OP first stage by deferring identification of all input coefficients to the second stage, using the proxy inversion solely to control for unobserved productivity. The ACF estimator uses the same variable definitions as the baseline: capx as the proxy variable, with output, labor, capital, and intermediate inputs defined identically to the OP specification.

³¹GDP Implicit Price Deflator: <https://fred.stlouisfed.org/series/A191RD3A086NBEA>.

³²Producer Price Index by Commodity: <https://fred.stlouisfed.org/series/WPUID61>.

³³Gross Private Domestic Investment Implicit Price Deflator: <https://fred.stlouisfed.org/series/A006RD3A086NBEA>.

Robustness: Gandhi, Navarro, and Rivers (2020) (GNR). The second robustness check follows [Gandhi et al. \(2020\)](#), who identify the output elasticity of the flexible input non-parametrically from the firm’s static optimality condition, without imposing functional form restrictions on the production function. In this framework, the proxy variable is the intermediate input share $\frac{m_{it}}{y_{it}}$ (cogs/sale), which under the firm’s cost-minimization condition is a sufficient statistic for productivity. The GNR estimator thus relaxes the parametric assumptions imposed by both OP and ACF, providing a more flexible benchmark for assessing the robustness of the output elasticity estimates.

A.2.2. Markup Estimation

Following [De Loecker et al. \(2020\)](#), firm-level markups are recovered from the firm’s cost-minimization condition for the variable input. For firm i in year t , the markup is defined as

$$\mu_{it} = \frac{\theta_{it}^m}{\alpha_{it}^m}, \quad (41)$$

where θ_{it}^m is the output elasticity of the variable input (intermediate inputs, cogs), recovered from the first-stage production function estimation, and $\alpha_{it}^m = \frac{P_{it}^m M_{it}}{P_{it} Y_{it}}$ is the expenditure share of intermediate inputs in total revenue. This approach requires no assumption on demand or market structure and yields a firm-year-specific markup that captures heterogeneity in both technology and market power. Markup estimates below 0.5 and above 10 are excluded as economically inadmissible, being inconsistent with profit-maximizing behavior and likely reflecting numerical instability in the cost share estimation. This restriction affects approximately 0.75% of the estimation sample.

A.3. Additional Figures and Empirical Results

Table A2. Summary Statistics

Panel A: Compustat Fundamentals <i>Obs.: 71,139 Firms: 8,125</i>					
Variable	Mean	SD	P25	Median	P75
Sale	3,398,985	17,129,640	22,203	121,055	846,755
Emp	9,814	37,290	135	597	3,702
AT	4,502,256	23,855,832	26,457	142,066	981,467
XRD	147,233	746,922	1,589	7,513	35,977
XSGA	641,985	2,651,924	11,640	44,070	208,185
Productivity (OLS)	16.916	22.373	5.118	7.423	18.825
Markup (OLS)	1.494	1.016	0.959	1.175	1.612
Productivity (ACF)	17.505	22.460	6.746	8.866	19.056
Markup (ACF)	1.219	0.761	0.832	1.019	1.341
Productivity (GNR)	12.536	28.759	0.416	0.801	2.441
Markup (GNR)	1.281	0.906	0.822	1.010	1.374
Panel B: Compustat-Segment Merged (including ADR) <i>Obs.: 43,007 Firms: 5,830</i>					
Variable	Mean	SD	P25	Median	P75
Sale	3,597,348	16,753,747	29,041	166,258	1,120,375
Emp	10,466	37,694	154	708	4,482
AT	5,101,380	26,098,750	37,743	211,989	1,398,987
XRD	180,872	867,769	2,154	10,697	49,903
XSGA	721,384	2,891,804	14,998	59,082	266,119
Productivity (OLS)	16.984	22.775	5.102	7.386	18.967
Markup (OLS)	1.501	1.029	0.957	1.171	1.623
Productivity (ACF)	17.726	23.397	6.718	8.758	18.886
Markup (ACF)	1.233	0.778	0.836	1.021	1.350
Productivity (GNR)	12.232	28.533	0.420	0.771	2.394
Markup (GNR)	1.290	0.926	0.822	1.008	1.379
Panel C: Compustat-Segment Merged (excluding ADR) <i>Obs.: 35,964 Firms: 4,905</i>					
Variable	Mean	SD	P25	Median	P75
Sale	2,171,465	10,468,548	25,104	135,367	826,899
Emp	6,440	23,946	133	571	3,212
AT	3,083,646	19,749,630	30,984	167,030	977,718
XRD	127,011	736,443	1,813	8,868	39,625
XSGA	476,449	2,256,258	13,128	50,168	208,927
Productivity (OLS)	16.551	22.397	5.066	7.262	17.726
Markup (OLS)	1.498	1.022	0.957	1.173	1.622
Productivity (ACF)	17.545	23.248	6.667	8.690	18.058
Markup (ACF)	1.236	0.778	0.839	1.027	1.357
Productivity (GNR)	11.965	28.419	0.414	0.732	2.203
Markup (GNR)	1.289	0.918	0.823	1.011	1.382

Notes: This table reports summary statistics for the three estimation samples. Panel A covers Compustat Fundamentals Annual restricted to firm-years with positive sales, employment, R&D (XRD), and SG&A (XSGA). Panel B merges Panel A with the Compustat Segment database, retaining all countries. Panel C restricts Panel B to U.S.-incorporated firms (*fic* = USA). All variables are in logarithms.

Table A3. Sector-Level Input Elasticity Estimates: OP, ACF, and GNR

NAICS	Sector	OP				ACF				GNR			
		β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS
11	Agriculture, Forestry & Fishing	0.818	0.038	0.107	0.964	0.405	0.222	0.253	0.881	0.665	0.200	0.147	1.012
21	Mining & Oil and Gas Extraction	0.714	0.120	0.054	0.888	0.793	0.159	0.088	1.040	0.495	0.377	0.113	0.984
23	Construction	0.924	0.035	0.008	0.967	0.688	0.031	0.187	0.906	0.815	0.021	0.135	0.971
31	Food & Beverage Manufacturing	0.815	0.064	0.082	0.961	—	—	—	—	0.662	0.174	0.177	1.013
32	Chemical & Paper Manufacturing	0.574	0.134	0.207	0.915	0.407	0.187	0.350	0.943	0.479	0.125	0.404	1.008
33	Machinery & Computer Manufacturing	0.711	0.079	0.191	0.981	0.653	0.117	0.189	0.959	0.618	0.084	0.342	1.044
42	Wholesale Trade	0.827	0.037	0.074	0.937	0.581	0.090	0.243	0.915	0.792	0.029	0.183	1.003
44	Retail Trade I	0.747	0.055	0.100	0.902	—	—	—	—	0.720	0.119	0.199	1.038
45	Retail Trade II	0.762	0.068	0.080	0.910	0.510	0.117	0.255	0.881	0.684	0.125	0.235	1.044
48	Transportation & Warehousing	—	—	—	—	0.594	0.147	0.108	0.850	0.758	0.208	0.047	1.013
51	Information	0.464	0.098	0.308	0.869	—	—	—	—	0.342	0.086	0.537	0.965
53	Real Estate & Rental	—	—	—	—	—	—	—	—	0.369	0.241	0.256	0.866
54	Professional & Technical Services	0.604	0.061	0.195	0.860	0.232	0.172	0.476	0.880	0.569	0.093	0.340	1.002
56	Administrative & Support Services	0.726	0.086	0.075	0.887	—	—	—	—	0.664	0.189	0.145	0.997
61	Educational Services	0.675	0.130	0.095	0.900	0.412	0.160	0.292	0.864	0.486	0.201	0.270	0.957
62	Health Care & Social Assistance	0.793	0.076	0.014	0.884	0.472	0.164	0.233	0.869	0.711	0.132	0.112	0.955
71	Arts, Entertainment & Recreation	0.770	0.094	0.018	0.881	0.409	0.320	0.109	0.838	0.646	0.212	0.071	0.929
72	Accommodation & Food Services	0.756	0.126	0.021	0.903	0.542	0.213	0.166	0.921	0.734	0.201	0.089	1.024
81	Other Services	0.796	0.009	0.110	0.915	0.263	0.182	0.399	0.844	—	—	—	—
99	Nonclassifiable Establishments	0.712	0.028	0.162	0.901	0.566	0.079	0.253	0.898	0.599	0.068	0.315	0.982

Notes: β_M , β_K , and β_L denote the estimated output elasticities with respect to materials (flexible input), capital, and labor (both dynamic inputs), respectively. RTS $\equiv \beta_M + \beta_K + \beta_L$ is the implied returns to scale. OP refers to olleypakes1996; ACF refers to ackerberg2015identification; GNR refers to gandhi2020identification. Missing values (—) indicate that the estimation did not produce valid positive elasticities or that the sector was not identified under the respective method.

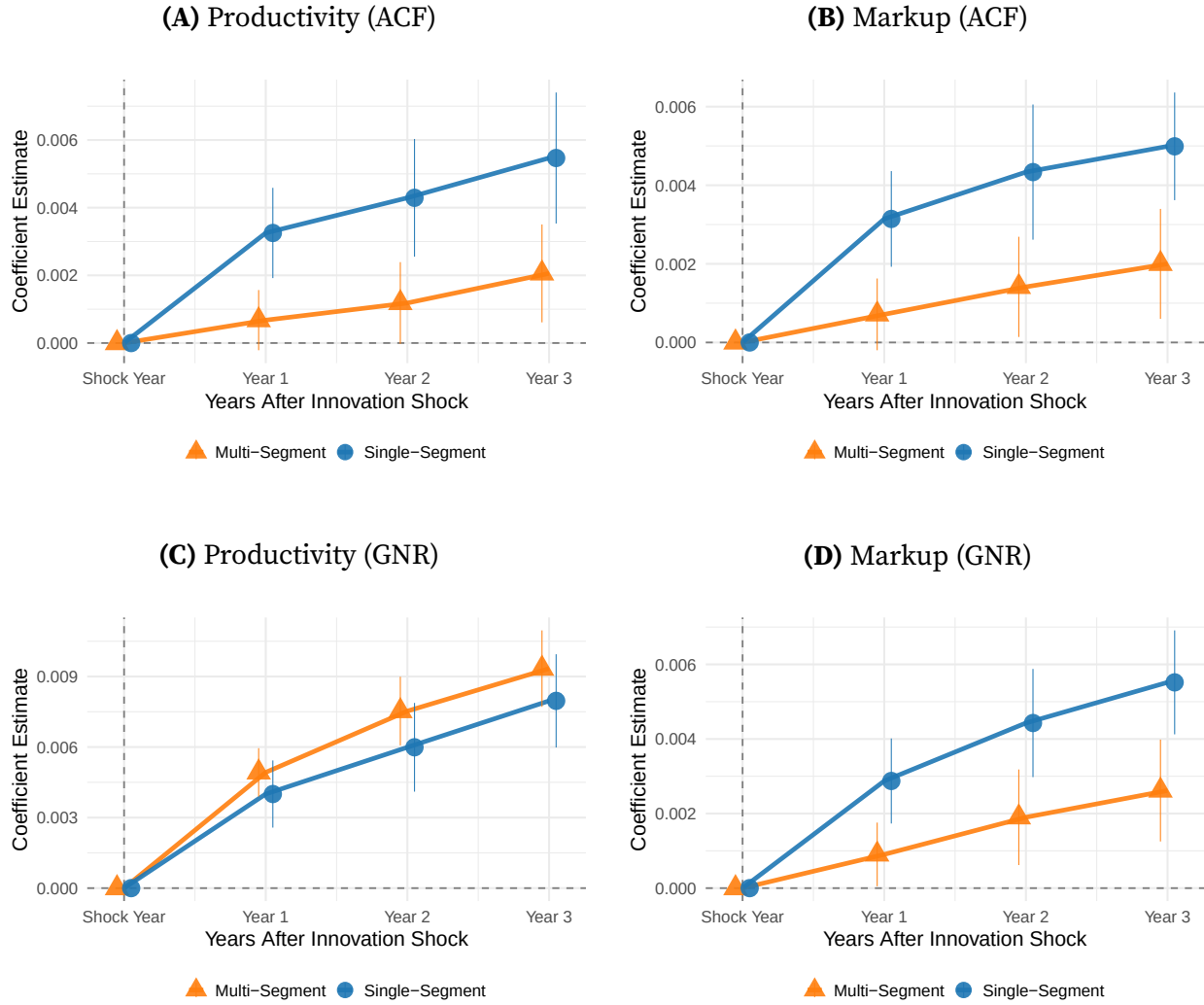


Figure A1. Productivity and Markup Responses to Innovation Shocks: ACF and GNR

Note: Panels (A) and (B) report estimates based on [Akerberg et al. \(2015\)](#); Panels (C) and (D) report estimates based on [Gandhi et al. \(2020\)](#). The innovation shock is measured as $\log(1 + \text{PatentValue}_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Shaded bands denote 95% confidence intervals based on standard errors clustered at the industry level.

Table A4. Regression Results
Panel A: Markup and Productivity

	Pooled OLS		Two-way FE	
	(2) <i>Markup</i>	(3) <i>Productivity</i>	(6) <i>Markup</i>	(7) <i>Productivity</i>
Production Lines	−0.035*** (0.003)	−0.067*** (0.009)	−0.038*** (0.004)	−0.045*** (0.004)
Num. Obs.	39,633	40,035	39,633	40,035
Adj. R^2	0.279	0.148	0.292	0.849
Covariates	Yes	Yes	Yes	Yes
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Panel B: Investment Ratio and Fluidity

	Pooled OLS		Two-way FE	
	(1) <i>Transferable/Embedded</i>	(4) <i>Comp. Threat</i>	(5) <i>Transferable/Embedded</i>	(8) <i>Comp. Threat</i>
Production Lines	−0.011*** (0.002)	−0.037*** (0.006)	−0.014*** (0.002)	−0.017** (0.006)
Num. Obs.	42,831	31,318	42,831	31,318
Adj. R^2	0.977	0.208	0.978	0.305
Covariates	Yes	Yes	Yes	Yes
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors are clustered by *firm id* in parentheses. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns 1–4: Pooled OLS specifications; Columns 5–8: Two-way fixed effects (year and industry).

B. Model Appendix

B.1. Final Good Sector Demand

The final good sector's profit maximization problem is:

$$\max_{y_{jt}} \exp \left(\int_0^1 \ln y_{jt} dj \right) - \int_0^1 p_{jt} y_{jt} dj. \quad (42)$$

The first-order condition yields the inverse demand function for each intermediate good j :

$$p_{jt} = \frac{Y_t}{y_{jt}}. \quad (43)$$

B.2. Intermediate Good Sector Demand

The cost minimization problem for the intermediate sector j is:

$$\min_{y_{sjt}, y_{fjt}} p_{sjt} y_{sjt} + p_{fjt} y_{fjt} \quad s.t. \quad y_{jt} = \left(\Xi(e_s) y_{sjt}^\varepsilon + (1 - \Xi(e_s)) y_{fjt}^\varepsilon \right)^{\frac{1}{\varepsilon}}. \quad (44)$$

The first-order condition with respect to the superstar firm's output y_{sjt} is:

$$p_{sjt} = \lambda \Xi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon-1}, \quad (45)$$

where λ is the Lagrange multiplier. Raising both sides to the power of $\frac{\varepsilon}{\varepsilon-1}$ and simplifying allows to solve for the ideal price index p_j for sector j :

$$\lambda = \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s))^{\frac{-1}{\varepsilon-1}} p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{\frac{\varepsilon-1}{\varepsilon}} \equiv p_j. \quad (46)$$

Substituting $\lambda = p_j$ back into the first-order condition yields the inverse demand function faced by the superstar firm:

$$p_{sjt} = p_j \Xi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon-1}. \quad (47)$$

Finally, substituting the final good producer's demand $y_{jt} = Y_t/p_j$ and solving for $y_{s jt}$ provides demand function:

$$y_{s jt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \Xi(e_s)^{\frac{1}{1-\varepsilon}} p_{s jt}^{\frac{1}{\varepsilon-1}} Y_t. \quad (48)$$

B.3. Superstar Firm Maximization Problem: Bertrand Competition

The superstar firm competes à la Bertrand with a continuum of fringe firms. Its profit maximization problem in industry j is:

$$\max_{p_{s jt}} (p_{s jt} - MC_{s jt}) y_{s jt} \quad s.t. \quad y_{s jt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \Xi(e_s)^{\frac{1}{1-\varepsilon}} p_{s jt}^{\frac{1}{\varepsilon-1}} Y_t. \quad (49)$$

Substituting the expression for p_j into the demand function, the objective function can be expanded as:

$$\max_{p_{s jt}} \left[p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} \Xi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) p_{f jt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right] \quad (50)$$

$$- MC_{s jt} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) p_{f jt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} p_{s jt}^{\frac{1}{\varepsilon-1}} \Xi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \right]. \quad (51)$$

After computing the derivative and factoring common terms, this condition can be expressed as:

$$\begin{aligned} \frac{\partial \pi}{\partial p_{s jt}} = & Y_t \Xi(e_s)^{\frac{1}{1-\varepsilon}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) p_{f jt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \times \left\{ \left[\frac{\varepsilon}{\varepsilon-1} p_{s jt}^{\frac{1}{\varepsilon-1}} - MC_{s jt} \frac{1}{\varepsilon-1} p_{s jt}^{\frac{2-\varepsilon}{\varepsilon-1}} \right] \right. \\ & \left. - \left(p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{s jt} p_{s jt}^{\frac{1}{\varepsilon-1}} \right) \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon-1} p_{s jt}^{\frac{1}{\varepsilon-1}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{s jt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) p_{f jt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right) \right\}. \end{aligned} \quad (52)$$

$$0 = \left[\frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{1}{\varepsilon-1}} - MC_{sjt} \frac{1}{\varepsilon - 1} p_{sjt}^{\frac{2-\varepsilon}{\varepsilon-1}} \right] - \left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{1}{\varepsilon-1}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right). \quad (53)$$

To simplify, multiply both sides by p_{sjt} and substitute the market share definition ϕ_{sjt} from (19), yielding:

$$\left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\phi_{sjt} \frac{\varepsilon}{\varepsilon - 1} \right) = \left[\frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{sjt} \frac{1}{\varepsilon - 1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \right]. \quad (54)$$

First, divide both sides by $p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}$, then by $\frac{\varepsilon}{\varepsilon-1}$. After rearranging terms, the expression simplifies to:

$$\varepsilon(1 - \phi_{sjt}) = \frac{MC_{sjt}}{p_{sjt}} (1 - \varepsilon \phi_{sjt}). \quad (55)$$

Solving for the optimal price p_{sjt} gives:

$$p_{sjt} = \frac{1 - \varepsilon \phi_{sjt}}{\varepsilon(1 - \phi_{sjt})} \cdot MC_{sjt}. \quad (56)$$

To determine the optimal labor demand, equating output (6) and demand (48) yields

$$q_{sjt} \psi(e_s, n_s) l_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \quad (57)$$

Multiplying both sides by p_{sjt} and dividing by w_t gives

$$p_{sjt} \underbrace{\frac{q_{sjt} \psi(e_s, n_s)}{w_t}}_{\text{inverse } MC_{sjt}} l_{sjt} = \underbrace{p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}}}_{\phi_{sjt}} \underbrace{\frac{Y_t}{w_t}}_{\omega_t^{-1}}. \quad (58)$$

This expression leads directly to equation (24).

B.4. Superstar Firm Maximization Problem: Cournot Competition

In the à la Cournot setup, a superstar firm and a continuum of fringe firms compete by choosing quantities to sell rather than engaging in price competition as in the à la

Bertrand case. Its profit-maximization problem in industry j is

$$\max_{y_{sjt}} \pi_{sjt} = \max_{y_{sjt}} (p_{sjt} - MC_{sjt}) y_{sjt}, \quad (59)$$

$$s.t. \quad p_{sjt} = \chi(e_s) y_{jt}^{-\varepsilon} y_{sjt}^{\varepsilon-1} Y_t, \quad \text{and} \quad y_{jt} = (\chi(e_s) y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon})^{1/\varepsilon}. \quad (60)$$

Differentiate the profit function with respect to y_{sjt} :

$$\frac{d\pi_{sjt}}{dy_{sjt}} = (p_{sjt} - MC_{sjt}) + y_{sjt} \frac{dp_{sjt}}{dy_{sjt}} = 0. \quad (61)$$

Differentiate the inverse demand (60) to obtain

$$\frac{dp_{sjt}}{dy_{sjt}} = p_{sjt} \left(-\varepsilon \frac{1}{y_{jt}} \frac{dy_{jt}}{dy_{sjt}} + \frac{\varepsilon - 1}{y_{sjt}} \right). \quad (62)$$

Differentiate y_{jt} with respect to y_{sjt} :

$$\frac{dy_{jt}}{dy_{sjt}} = \chi(e_s) y_{sjt}^{\varepsilon-1} y_{jt}^{1-\varepsilon}. \quad (63)$$

Substituting (63) and (60) into (61) yields

$$(p_{sjt} - MC_{sjt}) + y_{sjt} p_{sjt} \left(-\varepsilon \chi(e_s) y_{sjt}^{\varepsilon-1} y_{jt}^{-\varepsilon} + \frac{\varepsilon - 1}{y_{sjt}} \right) = 0. \quad (64)$$

Using the market-share definition

$$\frac{p_{sjt} y_{sjt}}{Y_t} = \chi(e_s) y_{sjt}^{\varepsilon} y_{jt}^{-\varepsilon},$$

and dividing both sides by p_{sjt} , rearrangement gives the inverse markup condition

$$\frac{MC_{sjt}}{p_{sjt}} = \varepsilon(1 - \phi_{sjt}). \quad (65)$$

The relative price ratio between fringe and superstar firms is therefore

$$\frac{p_{fjt}}{p_{sjt}} = \frac{1 - \phi_{sjt}}{\phi_{sjt}} \frac{MC_{fjt}}{MC_{sjt}}. \quad (66)$$

Using the inverse demand expressions leads to

$$\chi(e_s) \left(\frac{y_{fjt}}{y_{sjt}} \right)^{\varepsilon-1} = \frac{1 - \phi_{sjt}}{\phi_{sjt}} \frac{MC_{fjt}}{MC_{sjt}}. \quad (67)$$

Substituting the marginal-cost expressions for the fringe and superstar firms yields

$$\left(\frac{y_{fjt}}{y_{sjt}} \right)^{\varepsilon-1} = \frac{1}{\chi(e_s)} \frac{1 - \phi_{sjt}}{\phi_{sjt}} \frac{1}{\lambda^{m_j} \psi(e_s, n_s)}. \quad (68)$$

Equation (68) shows that the relative output depends on the market share, the quality gap, the embedded intangible level, and the firm's production-line parameter. Rearranging the market-share definition gives

$$\chi(e_s) y_{sjt}^{\varepsilon} y_{jt}^{-\varepsilon} = \frac{\chi(e_s) y_{sjt}^{\varepsilon}}{\chi(e_s) y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}} = \frac{1}{1 + \frac{1}{\chi(e_s)} \left(\frac{y_{fjt}}{y_{sjt}} \right)^{\varepsilon}}, \quad (69)$$

which shows that market share depends on the output gap and the embedded intangible level. Therefore, the output gap depends on the quality gap, the embedded intangible level, and the number of production lines associated with superstar s .

B.5. Aggregate Output and Growth Rate

Using the superstar (6) and fringe firm (7) output equations into (3) gives

$$Y_t = \exp \left(\int_0^1 \ln \left[(\xi e_{st})^{\beta} \left(q_{sjt} \frac{((1 - \xi) e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}} l_{sjt} \right)^{\varepsilon} + (q_{fjt} l_{fst})^{\varepsilon} \right]^{1/\varepsilon} dj \right)$$

$$Y_t = \exp \left(\int_0^1 \ln \left[\left(q_{sjt}^{\varepsilon} \left((\xi e_{st})^{\beta} \left(\frac{((1 - \xi) e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}} l_{sjt} \right)^{\varepsilon} + (\lambda^{-m_{jt}} l_{fst})^{\varepsilon} \right) \right) \right]^{1/\varepsilon} dj \right)$$

$$= \underbrace{\exp \left(\int_0^1 \ln q_{sjt} dj \right)}_{Q_t} \exp \left(\int_0^1 \ln \left[((\xi e_{st})^\beta \left(\frac{((1-\xi)e_{st})^\alpha}{\gamma n_{st}^{\alpha_s}} l_{sjt} \right)^\varepsilon + (\lambda^{-m_{jt}} l_{fst})^\varepsilon)^\frac{1}{\varepsilon} dj \right] \right), \quad (70)$$

and , along with the labor demands in (24) and (??), and factoring out ω_t^{-1} , aggregate output can be expressed as

$$Y_t = Q_t \omega_t^{-1} \exp \left(\int_0^1 \ln \left[(\xi e_{st})^\beta \left(\frac{((1-\xi)e_{st})^\alpha \phi_{sjt}}{\gamma n_{st}^{\alpha_s} \sigma_{sjt}} \right)^\varepsilon + (\lambda^{-m_{jt}} (1 - \phi_{sjt}))^\varepsilon \right]^\frac{1}{\varepsilon} dj \right). \quad (71)$$

Because everything in the integrand depends only on the gaps (m, k, n) , the expression can be written in discrete state space as

$$Y_t = Q_t \omega_t^{-1} \exp \left(\underbrace{\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \ln \left[(\xi \theta^k)^\beta \left(\frac{((1-\xi)\theta^{kt})^\alpha \phi_t(m, k, n)}{\gamma n_{st}^{\alpha_s} \sigma_t(m, k, n)} \right)^\varepsilon + (\lambda^{-mt} (1 - \phi_t(m, k, n)))^\varepsilon \right]^\frac{1}{\varepsilon}}_{\equiv R_t(m, k, n)} \mu_t(m, k, n) \right). \quad (72)$$

$$\begin{aligned} \ln Y_{t+\Delta t} - \ln Y_t &= (\ln Q_{t+\Delta t} - \ln Q_t) + \ln \omega_t - \ln \omega_{t+\Delta t} \\ &\quad + \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(R_{t+\Delta t}(m, k, n) - R_t(m, k, n) \right) \left(\mu_{t+\Delta t}(m, k, n) - \mu_t(m, k, n) \right) + o(\Delta t). \end{aligned} \quad (73)$$

where

$$\begin{aligned} \ln Q_{t+\Delta t} - \ln Q_t &= \ln \lambda \left[\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(z_t^{Int}(m, k, n) + p_{s \geq s'}^{Ex} z_t^{Ex}(m, k, n) + Z_t^f(m, k, n) \right) \mu_t(m, k, n) \right] \Delta t \\ &\quad + o(\Delta t). \end{aligned} \quad (74)$$

Dividing by Δt and taking the limit $\Delta t \rightarrow 0$, the growth rate of the economy is

$$g_t = -g_{\omega,t} + g_{Q,t} + g_{R,t}. \quad (75)$$

In the steady state, the distribution $\mu_t(m, k, n)$ is constant, implying that R_t is constant. Wages grow at the same rate as output, so the real wage remains constant. Therefore, in steady state the growth rate of the economy is determined solely by quality improvements:

$$g = g_Q = \ln \lambda \left[\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(z^{\text{Int}}(m, k, n) + p_{s \geq s'}^{\text{Ex}} z^{\text{Ex}}(m, k, n) + Z^f(m, k, n) \right) \mu(m, k, n) \right]. \quad (76)$$

B.6. Decomposition of Output

Starting from (71) and factoring out the term $\Xi(e_{st}) \left(\frac{((1-\xi)e_{st})^\alpha}{\gamma n_{st}^{\alpha_s}} \right)$, aggregate output can be written as

$$Y_t = Q_t \omega_t^{-1} E_t \exp \left(\int_0^1 \ln \left[\left(\frac{\phi_{sjt}}{\sigma_{sjt}} \right)^\varepsilon + \frac{1}{(\xi e_{st})^\beta} \left(\frac{\gamma n_{st}^{\alpha_s}}{((1-\xi)e_{st})^\alpha} \lambda^{-m_{jt}} (1 - \phi_{sjt}) \right)^\varepsilon \right]^{1/\varepsilon} dj \right). \quad (77)$$

Define the multiplicative factor that collects the factored-out terms as

$$E_t = \exp \left(\int_0^1 \ln \left[\Xi(e_{st}) \left(\frac{((1-\xi)e_{st})^\alpha}{\gamma n_{st}^{\alpha_s}} \right)^\varepsilon \right]^{1/\varepsilon} dj \right). \quad (78)$$

Next multiply and divide the integrand by the linear weight $\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt})$. After this algebraic step I obtain a decomposition that isolates a simple mean term and a residual term:

$$Y_t = Q_t E_t \omega_t^{-1} \exp \left(\int_0^1 \ln \left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt}) \right] dj \right) \times \exp \left(\int_0^1 \ln \left[\frac{\left(\frac{1}{\gamma n_{st}^{\alpha_s} \sigma_{sjt}} \right)^\varepsilon + \frac{1}{(\xi e_{st})^\beta} \left(\frac{1}{((1-\xi)e_{st})^\alpha} \lambda^{-m_{jt}} (1 - \phi_{sjt}) \right)^\varepsilon}{\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt})} \right]^{1/\varepsilon} dj \right). \quad (79)$$

Finally, using (34) and defining the multiplicative mean term

$$M_t = \frac{\exp\left(\int_0^1 \ln \left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt}) \right] dj\right)}{\int_0^1 \left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt}) \right] dj}, \quad (80)$$

the output decomposition can be written as

$$Y_t = Q_t \times E_t \times M_t \times S_t,$$

where

$$S_t = \exp \left(\int_0^1 \ln \left[\frac{\left(\frac{1}{\gamma n_{st}^{\alpha_s} \sigma_{sjt}} \right)^\varepsilon + \frac{1}{(\xi e_{st})^\beta} \left(\frac{1}{((1 - \xi) e_{st})^\alpha} \lambda^{-m_{jt}} (1 - \phi_{sjt}) \right)^\varepsilon}{\frac{\phi_{sjt}}{\sigma_{sjt}} + (1 - \phi_{sjt})} \right]^{1/\varepsilon} dj \right). \quad (81)$$

B.7. Consumption Equivalence Welfare Measure

On the balanced growth path, consumption grows at rate g , so that $C(t) = C_0 \exp(gt)$.³⁴

Defining welfare Ω as the present value of lifetime utility from consumption yields:

$$\Omega = \int_0^\infty e^{-\rho t} \ln(C(t)) dt \quad (83)$$

$$= \ln(C_0) \int_0^\infty e^{-\rho t} dt + g \int_0^\infty t e^{-\rho t} dt. \quad (84)$$

³⁴The C_0 consumption level is given by:

$$C_0 = Y_0 - \int_0^1 \left(I_0^{\text{Int}} + I_0^{\text{Emb}} + I_0^{\text{Ex}} + I_0^f \right) dj + G_0. \quad (82)$$

In this equation, the subscript 0 represents calibrated optimum values on the balanced growth path, and G is the lump-sum transfer of government taxes.

Solving these integrals gives:

$$\Omega = \frac{\ln(C_0)}{\rho} + \frac{g}{\rho^2} \quad (85)$$

$$= \frac{1}{\rho} \left(\ln C_0 + \frac{g}{\rho} \right). \quad (86)$$

Equivalent Welfare Changes Between Economies To compare welfare between two economies—a calibrated benchmark economy (Cal) and a taxed economy (Tax) on their respective balanced growth paths—I compute the percentage change δ in lifetime consumption that would make households indifferent between the two. The required compensation δ satisfies:

$$\Omega^{\text{Tax}} = \frac{1}{\rho} \left(\ln (C_0^{\text{Cal}}(1 + \delta)) + \frac{g^{\text{Cal}}}{\rho} \right). \quad (87)$$

Solving equation (87) for δ :

$$\frac{\ln C_0^{\text{Tax}}}{\rho} + \frac{g^{\text{Tax}}}{\rho^2} = \frac{\ln [C_0^{\text{Cal}}(1 + \delta)]}{\rho} + \frac{g^{\text{Cal}}}{\rho^2} \quad (88)$$

$$\ln \left(\frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}} \right) + \frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho} = \ln(1 + \delta) \quad (89)$$

$$\delta = \frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}} \exp \left(\frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho} \right) - 1. \quad (90)$$

If $\delta > 0$: households require compensation to remain in the benchmark economy (Cal).

If $\delta < 0$: households would pay to move to the taxed economy (Tax).

C. Numerical Appendix

C.1. Additional Numerical Results

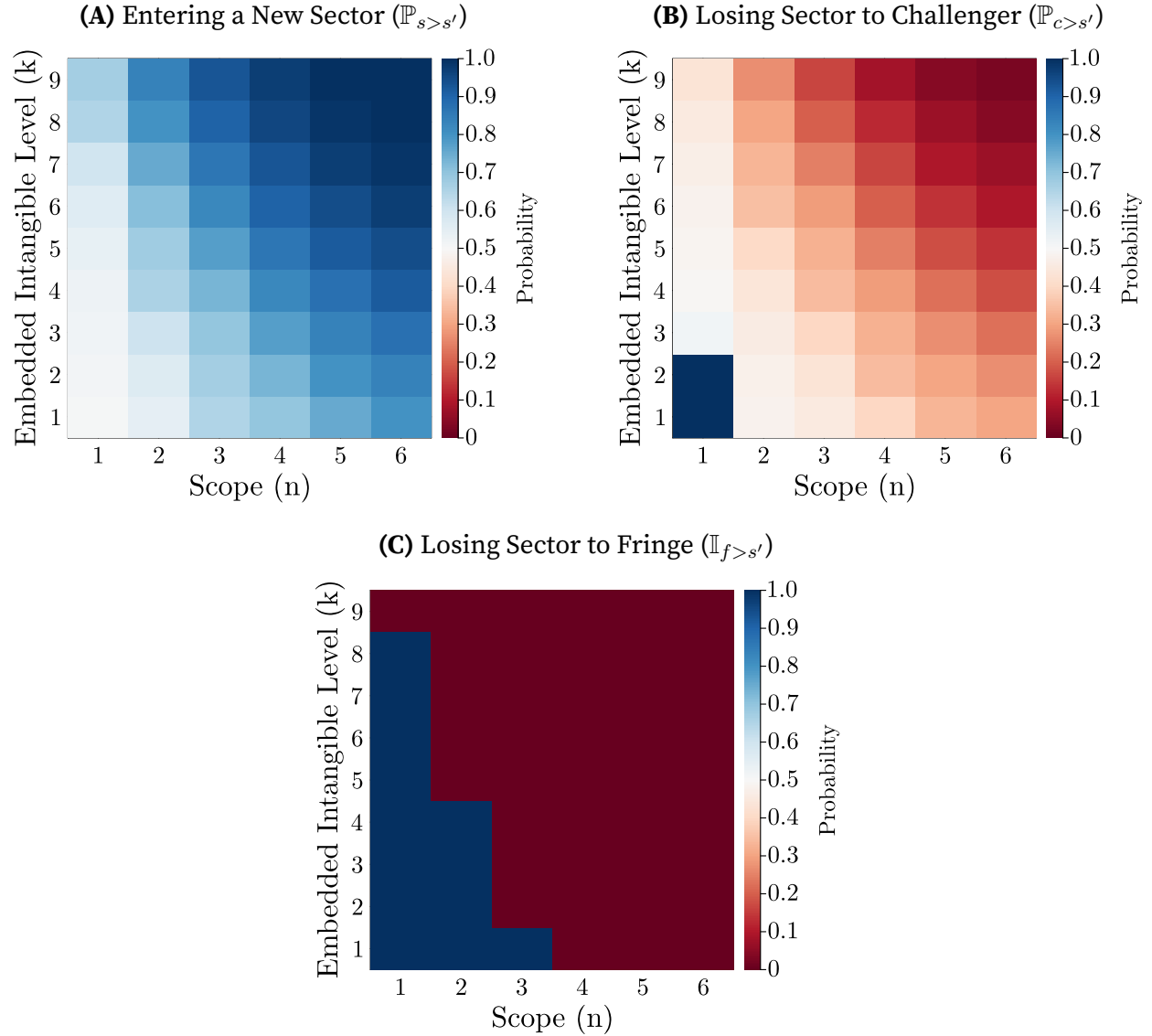


Figure C1. Superstar Firm Entry and Displacement Probabilities

Note: Each panel displays a probability evaluated over the (n, k) grid, where the horizontal axis reports firm scope n and the vertical axis reports the embedded intangible level k . Darker blue indicates higher probability and darker red indicates lower probability.

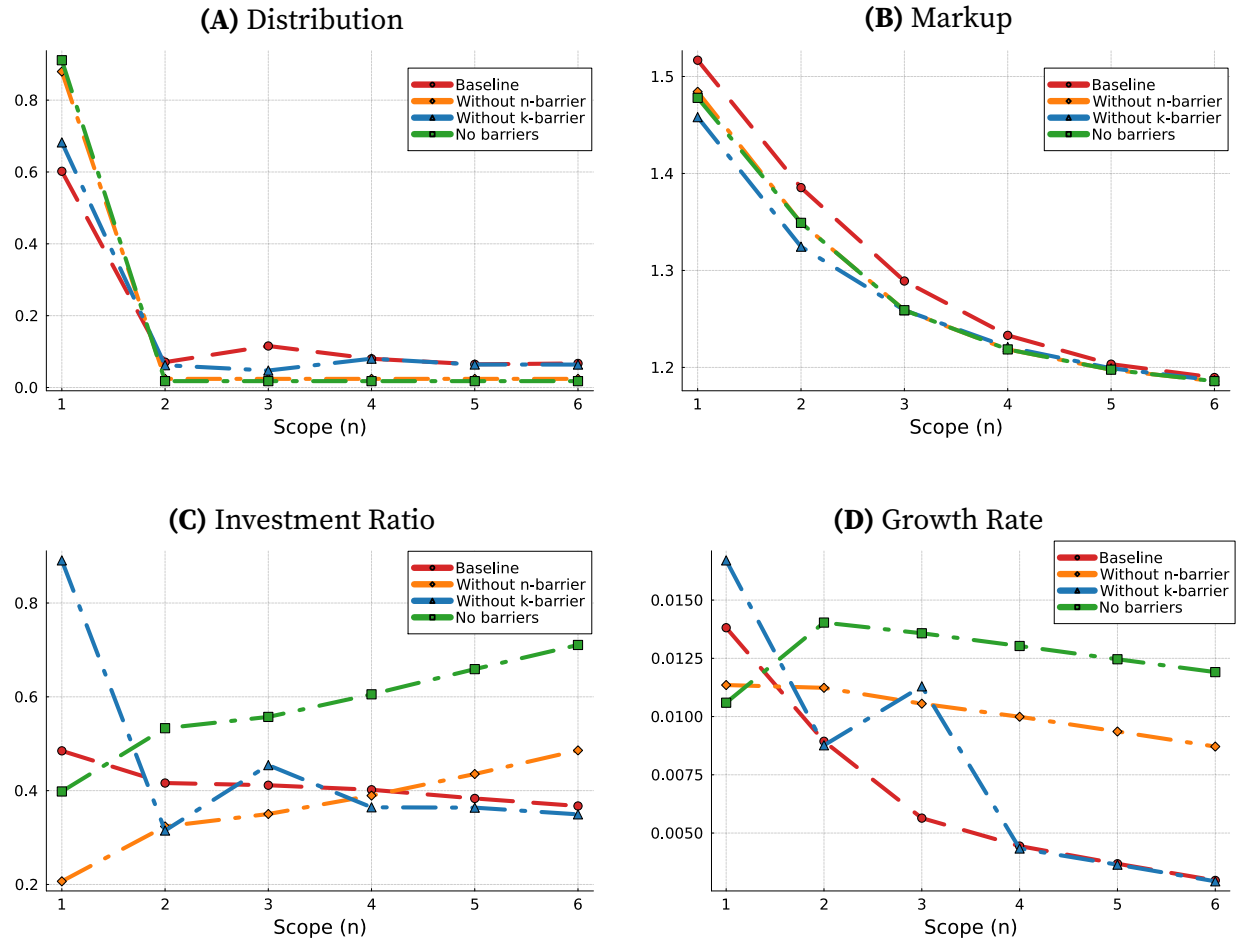


Figure C2. Removing Entry Barrier Scenarios

Note: The green line represents the internally calibrated optimal value of ξ , while the blue line shows an upward shift in ξ and the orange line shows a downward shift. The horizontal axis corresponds to the production line dimension.

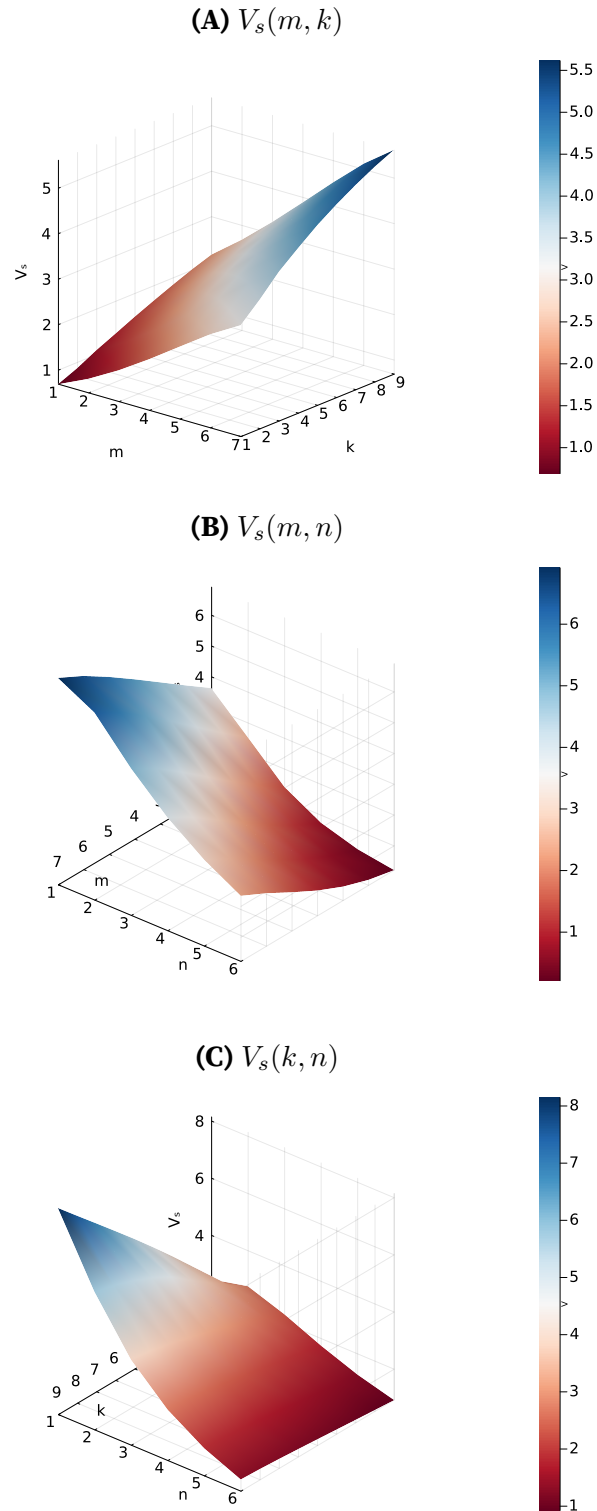


Figure C3. Superstar Value Function

Note: Each panel displays a probability evaluated over the (n, k) grid, where the horizontal axis reports firm scope n and the vertical axis reports the embedded intangible level k . Darker blue indicates higher probability and darker red indicates lower probability.

Table C1. Policy Experiments: Growth and Welfare Effects ($|\tau| = 0.2$)

Pattern	Without Subsidizing Fringe Firms				With Subsidizing Fringe Firms $\tau_f = 0.2$			
	$\Delta g\%$	$\Omega\%$	C_{new}/C_0	$\exp(\Delta g/\rho)$	$\Delta g\%$	$\Omega\%$	C_{new}/C_0	$\exp(\Delta g/\rho)$
Panel A: τ_{scope} only								
All +0.2	-25.946	-8.443	0.973	0.942	-11.395	2.425	1.052	0.974
First 1 = -0.2, rest +0.2	-1.595	10.396	1.108	0.996	12.934	22.675	1.191	1.030
First 2 = -0.2, rest +0.2	24.101	1.732	0.962	1.057	48.536	11.556	0.997	1.119
First 3 = -0.2, rest +0.2	28.496	2.011	0.955	1.068	53.596	11.189	0.982	1.132
First 4 = -0.2, rest +0.2	28.249	2.093	0.956	1.068	53.106	11.477	0.986	1.131
First 5 = -0.2, rest +0.2	28.409	2.211	0.957	1.068	53.093	11.586	0.987	1.131
All -0.2	29.060	2.677	0.960	1.070	53.684	12.063	0.990	1.133
Panel B: τ_{hor} only								
All +0.2	-2.446	1.034	1.016	0.994	16.217	14.178	1.100	1.038
First 1 = -0.2, rest +0.2	4.800	-0.865	0.980	1.011	26.887	10.904	1.042	1.064
First 2 = -0.2, rest +0.2	4.559	-1.372	0.976	1.011	26.457	10.273	1.037	1.063
First 3 = -0.2, rest +0.2	3.609	-1.903	0.973	1.008	25.262	9.680	1.034	1.060
First 4 = -0.2, rest +0.2	2.808	-2.376	0.970	1.006	24.306	9.171	1.032	1.058
First 5 = -0.2, rest +0.2	2.290	-2.668	0.968	1.005	23.701	8.870	1.031	1.057
All -0.2	2.290	-2.668	0.968	1.005	23.701	8.870	1.031	1.057
Panel C: τ_{emb} only								
All +0.2	-1.863	0.279	1.007	0.996	17.868	13.637	1.090	1.042
First 1 = -0.2, rest +0.2	-1.188	-1.656	0.986	0.997	18.677	10.572	1.059	1.044
First 2 = -0.2, rest +0.2	-0.292	-1.434	0.986	0.999	19.766	10.778	1.058	1.047
First 3 = -0.2, rest +0.2	0.508	-1.227	0.987	1.001	20.736	10.978	1.058	1.049
First 4 = -0.2, rest +0.2	1.158	-1.080	0.987	1.003	21.512	11.110	1.057	1.051
First 5 = -0.2, rest +0.2	1.692	-0.972	0.986	1.004	22.151	11.202	1.056	1.053
First 6 = -0.2, rest +0.2	2.008	-0.935	0.986	1.005	22.516	11.224	1.056	1.054
First 7 = -0.2, rest +0.2	2.197	-0.923	0.986	1.005	22.740	11.231	1.055	1.054
First 8 = -0.2, rest +0.2	2.304	-0.903	0.986	1.005	22.873	11.240	1.055	1.054
All -0.2	2.304	-0.903	0.986	1.005	22.873	11.240	1.055	1.054

Table C2. Sensitivity Analysis: Effect of a $\pm 1\%$ Parameter Perturbation on Model Outcomes

Parameter	Investment Ratio		Markup		Growth Rate	
	+1%	−1%	+1%	−1%	+1%	−1%
<i>Demand and Technology</i>						
CES parameter (ε)	1.960	−1.447	−0.096	0.173	2.779	−2.559
Curvature of managerial quality (α)	1.305	127.472	0.171	−1.769	3.474	4.354
Curvature of brand value (β)	−0.062	0.063	−0.054	0.054	−0.065	0.066
Scale of managerial quality (γ)	−2.932	3.130	−0.423	0.455	−1.602	1.626
Share of brand value (ξ)	−0.586	0.581	−0.085	0.085	−0.287	0.298
Quality improvement step size (λ)	334.219	−23.373	3.839	−4.457	62.250	−34.154
Embedded innovation step size (θ)	14.919	112.033	0.284	−2.206	8.750	1.256
<i>Curvature and Cost Scale Parameters</i>						
Curvature of embedded innovation (ϑ_{Emb})	1.519	−1.532	0.016	−0.015	0.923	−0.935
Cost scale of internal innovation (γ^{Ver})	−0.422	0.431	0.008	−0.008	−0.202	0.206
Cost scale of horizontal innovation (γ^{Hor})	−0.291	0.297	0.032	−0.032	−0.102	0.110
Cost scale of embedded innovation (γ^{Emb})	−0.026	0.026	−0.002	0.002	−0.102	0.104
Cost scale of fringe (γ^f)	−0.493	0.504	−0.031	0.031	−0.819	0.835
Baseline value	1.3724		1.5731		0.0116	

Notes: Each entry reports the percentage change in the outcome when the row parameter is perturbed by +1% or −1%, holding all other parameters at their baseline values. The *Intangibility Ratio* is defined as $(G_{\text{ver}} + G_{\text{hor}} + G_f) / G_{\text{emb}}$, where $G_k = \gamma_k z_k^{\vartheta_k}$. The *Markup* is $(1 - \varepsilon) / [(1 - \varphi) \varepsilon] + 1$. The *Growth Rate* is $\mathbb{E}[\log \lambda \cdot (z_{\text{ver}} + p_{\text{ex}} z_{\text{hor}} + \bar{m} p_f z_f)]$. Asymmetry between the +1% and −1% columns reflects local nonlinearity of the equilibrium mapping.

C.2. Solution Algorithm

This algorithm computes the balanced growth path with a three dimensional state space (m, k, n) . The solution involves finding the value functions $v_s(m, k, n)$ and $v_f(m, k, n)$, the innovation rates $z^{\text{Int}}, z^{\text{Emb}}, z^{\text{Ex}}, z^f$, and the stationary distribution $\mu(m, k, n)$ that jointly satisfy the model's equilibrium conditions.

BGP Equilibrium Solution:

1. **Compute static values:** Calculate static market shares and profit values using equations (??) and (19).
2. **Initialization:** Initialize the value functions $v_s(m, k, n)$ and $v_f(m, k, n)$, and the stationary distribution $\mu(m, k, n)$.
3. **Step 1: Solve HJB Equations (Backward Iteration)**
 - (a) Set $v^{\text{old}}(m, k, n)$.
 - (b) Repeat until $\max |v^{\text{new}} - v^{\text{old}}| < \text{tolerance}$:
 - Compute policy functions $x(m, k, n)$ from the FOCs using v^{old} .
 - Solve the discretized HJB equations for $v^{\text{new}}(m, k, n)$.
 - Update $v^{\text{old}} \leftarrow v^{\text{new}}$.
4. **Step 2: Solve the Kolmogorov Forward Equation (KFE)**
 - (a) Set $\mu^{\text{old}}(m, k, n)$.
 - (b) Repeat until $\max |\mu^{\text{new}} - \mu^{\text{old}}| < \text{tolerance}$:
 - Solve the discretized KFE for $\mu^{\text{new}}(m, k, n)$ using the policy functions $z^{\text{Int}}, z^{\text{Emb}}, z^{\text{Ex}}, z^f$.
 - Update $\mu^{\text{old}} \leftarrow \mu^{\text{new}}$.
5. **Step 3: Repeat Steps Until Value Functions and Distribution Converge**

Finally, to determine the optimal parameter values, search over the parameter space to minimize the objective function,

$$\text{Minimize}(z) = \sum_{z=1}^Z \frac{|\text{model}(z) - \text{data}(z)|}{\frac{1}{2}|\text{model}(z)| + \frac{1}{2}|\text{data}(z)|}.$$